# ExoMind: Democratizing Scientific Intelligence via Extended-Mind-Inspired Agentic System

### *Surpassing Frontier Proprietary Models in Scientific Reasoning and Research with Less Data, Small Model, and Low-cost Training*

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## Abstract

Scientific intelligence, which requires long-horizon reasoning and research across diverse and highly specialized domains, remains a key challenge for artificial intelligence. Although large language models (LLMs) have achieved strong performance on general tasks, their single-model paradigm faces fundamental limitations in scientific domains, including reduced effectiveness under specialization, high scaling costs with diminishing returns, and insufficient capacity for fine-grained long-horizon tasks. In this paper, we introduce an extended-mind-inspired paradigm, extending intelligence to an agentic system composed of LLM, interactive objects, and autonomous interaction processes, enabling deep specialization, new scaling dimensions, and stronger representational capacity. Building on this, we introduce **ExoMind**, the first extended-mind-inspired agentic system, which integrate systematic data engineering, scientific interaction framework, and systematic training strategy. Across diverse scientific reasoning and research tasks, our method achieves substantial and consistent performance improvements using less data, small model, and low-cost training. Notably, **ExoMind** achieves leading performance among frontier open-source and closed-source models on challenging scientific benchmarks (as shown in Fig. 1), while also improving general-purpose capabilities. These results suggest a customizable, cost-efficient, and high-performance pathway toward democratizing scientific intelligence.

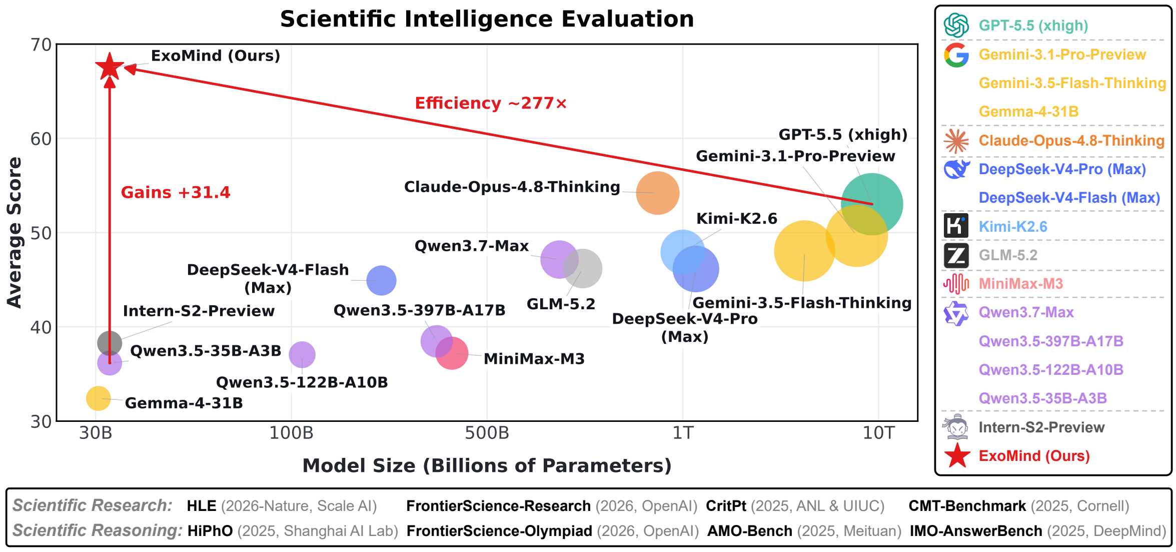


Figure 1 | **Our ExoMind surpassing frontier models on scientific research and reasoning benchmarks.** It achieves substantial gains over base model and attains the strongest average performance across eight benchmarks (see Tab. 2 and Fig. 8 for details), outperforming leading open- and closed-source models. Notably, these results are obtained with a compact 35B model, achieving ~ 277× parameter efficiency over GPT-5.5 (estimated at ~ 9.7 T parameters by IKP [1]). These findings highlight that we provide an efficient and effective path toward democratizing scientific intelligence.

# Contents

*"The Extended Mind Thesis: The mind is not confined to the brain, but extends into tools, artifacts, and the environment with which it interacts." – by Andy Clark and David Chalmers.*

# 1. Introduction

In recent years, artificial intelligence, particularly large language models (LLMs), has achieved remarkable progress across a wide range of general-purpose tasks, pushing many traditional benchmarks toward saturation. Scientific intelligence, characterized by cross-disciplinary knowledge structures, high domain barriers, and long-horizon complexity, has emerged as a critical frontier. Although LLMs demonstrate strong capabilities, this single-model paradigm still faces three fundamental limitations in scientific domains: (1) specialization often comes at the cost of generalization, and certain scientific tasks remaining difficult to model with only LLMs; (2) improvements that rely on larger models, more data, and longer training are increasingly costly and exhibit diminishing returns, especially in scientific fields; and (3) the averaged representations induced by a single LLM inevitably struggle to capture the fine-grained complexity of various scientific reasoning and research processes; as shown in Fig. 2. Therefore, advancing scientific intelligence beyond its bottlenecks requires not only further scaling but also new paradigms that enable superior and general capabilities under limited resources, thereby decoupling high-level intelligence from a small number of large-scale proprietary models and making them more accessible and reproducible.

In philosophy of mind, the Extended Mind Thesis says that the mind is not confined to the brain, but extends into objects and the environment with which it interacts, which serve as extensions of cognitive processes [2]. Inspired by this, we introduce **an extended-mind-inspired paradigm**: Instead of relying solely on the internal parameters and intrinsic capabilities of a single LLM, we reformulate intelligence as an agentic system composed of LLM, interactive objects, and autonomous interaction processes. Through long-horizon autonomous interactions with diverse objects, this system

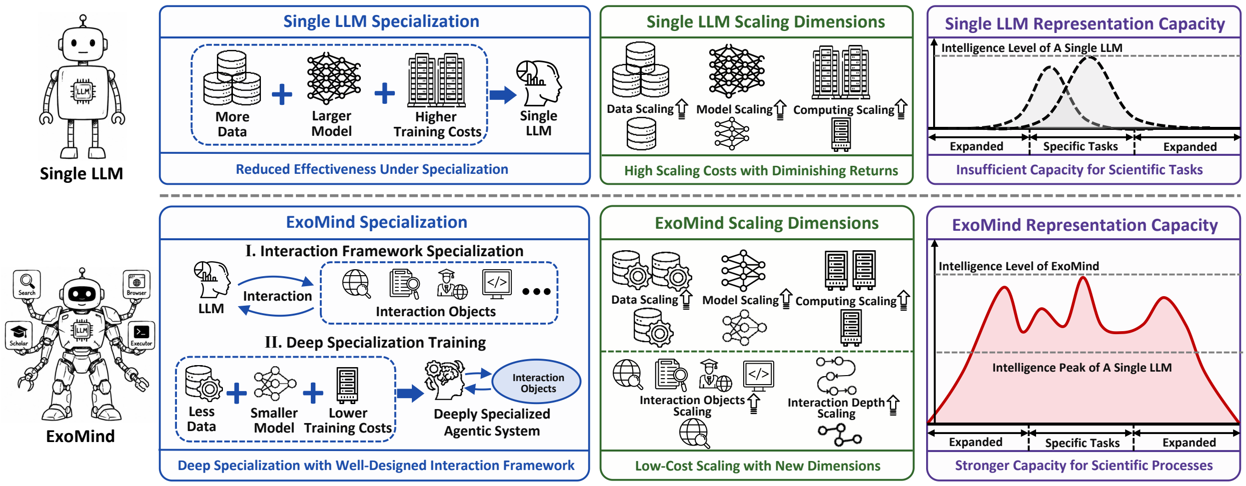


Figure 2 | **Paradigm comparison between single LLM and our ExoMind.** Single LLM paradigm achieves specialization by compressing heterogeneous tasks into one model and mainly scales through more data, larger model, and higher training costs, leading to reduced specialization effectiveness, costly diminishing returns, and limited representation capacity. Our ExoMind paradigm organizes intelligence as an agentic system with LLM, interaction objects, and autonomous interaction processes. It achieves specialization by interaction-framework specialization and deep-specialization training, and introduces new scaling dimensions over interaction objects and interaction depth, enabling superior and general capabilities under low-cost settings.

gives rise to emergent capabilities that surpass the boundaries of individual LLMs. It offers three key advantages for scientific intelligence: (1) **deep specialization**, where the focus of specialization can shift from modifying LLM parameters to designing external interactive objects, which can be easily tailored to kinds of tasks and domains; (2) **new scaling dimensions**, where capability growth is no longer limited to model size, data, and training time, but can be expanded through richer objects and deeper interactions; and (3) **stronger representational capacity**, an agentic intelligence system can better capture the diverse, fine-grained, and long-horizon characteristics of scientific process. Thus, it has the potential to break the trade-off among customizability, cost efficiency, and performance.

In this work, we further demonstrate that the innovative paradigm of intelligence organization can democratize scientific intelligence. Specifically, we introduce **ExoMind**, the first extended-mind-inspired agentic system designed for scientific domains. The construction process of our scientific agentic system consists of three main stages: (1) **a systematic data engineering pipeline**, which constructs a large-scale and diverse scientific data pool from different disciplines, and performs multi-stage fine-grained data filtering using the world-model-like predictive capabilities of strong models; (2) **a scientific interaction framework**, which defines atomic scientific capabilities and typed interaction objects for scientific reasoning and research, and introduces Chain-of-Interaction distillation technique combined with multilevel trajectory verification to efficiently enhance the long-horizon autonomous interaction abilities; and (3) **a systematic training strategy**, which enforces training-inference consistency constraint to fully unlock the potential of the scientific interaction framework, and employs a hybrid progressive CoI training scheme to enhance both the reasoning and interaction abilities, achieving superior scientific intelligence even under a low-cost condition of less data, small model, and low-cost training.

We systematically evaluate the proposed ExoMind on a broad range of scientific intelligence tasks. The evaluation spans various multi-disciplinary scientific reasoning and research datasets, covering problems of varying difficulty, as well as both open-set and closed-set settings. Our method achieves substantial performance improvements in all experimental settings at an extremely low cost, requiring only approximately 8 GPUs, 1–2 days of training, and a few thousand high-quality training trajectories. In particular, our ExoMind achieves new state-of-the-art performance among open-source models across all benchmarks, and approaches or surpasses frontier proprietary models in most benchmarks, as demonstrated in Fig. 1. Further, we evaluate our **ExoMind** on a diverse suite of general intelligence tasks, and find that it can achieve deep specialization while improving general intelligence. In addition, we conduct a series of interesting discussion experiments, including exploring the impact of training-inference consistency, the emergence of scientific interaction patterns, and the effects of scientific interaction framework. Overall, our method achieves a powerful performance using extremely-low cost, highlighting an efficient and effective pathway for democratizing scientific intelligence and accelerating artificial general intelligence.

## 2. Methods

### 2.1. Systematic Data Engineering Pipeline

To construct an extended-mind-inspired agentic system, we propose an efficient data filtering pipeline, as shown in Fig. 3. Specifically, we first build a large-scale and diverse scientific problem-answer pool for foundational scientific domains. Then, the world-model-like prediction capability of closed-source strong models is leveraged to evaluate candidate problems before generating trajectories, focusing on problem difficulty and potential tool benefit. Finally, through a multi-stage and fine-grained filtering process, we transform the raw problem-answer pool into a smaller but high-quality problem-answer pool and route each problem to an appropriate reasoning path. This process enables a compact set of high-quality data to provide effective and stable learning signals for both pure-reasoning

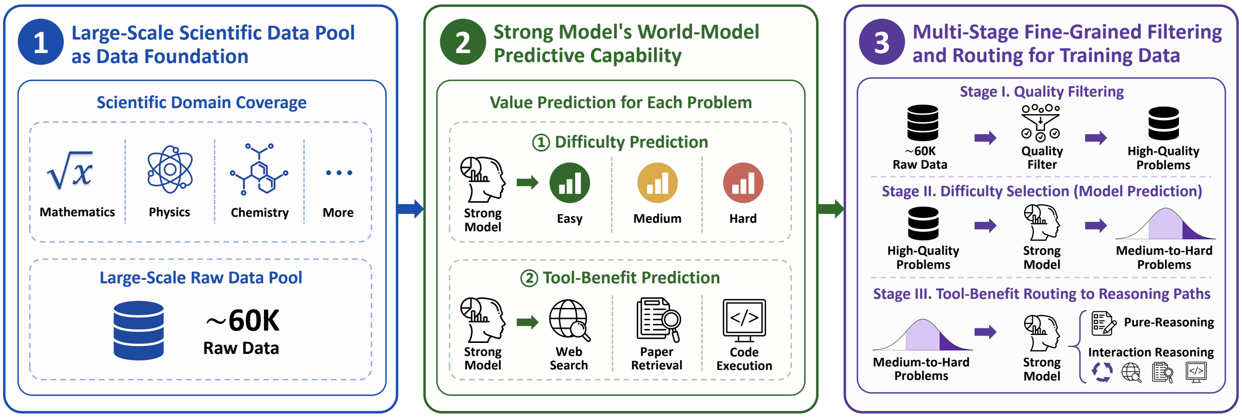


Figure 3 | Overview of the systematic data engineering pipeline. We construct a large-scale scientific data pool as the data foundation, use strong models to predict each problem's difficulty and tool benefit before trajectory generation, and progressively filter the pool through quality filtering, difficulty selection, and tool-benefit routing. The resulting problems provide a systematic source for generating both pure-reasoning and interaction-reasoning trajectories.

and interaction-reasoning, supporting the training of the proposed agentic system and laying the foundation for agentic scientific intelligence.

### 2.1.1. Large-Scale and Diverse Scientific Data Pool

We construct a large-scale and diverse scientific data pool from different disciplines as the foundation to advance scientific intelligence. In detail, we first construct an initial problem-answer pool of approximately 60K samples, covering foundational areas such as mathematics, physics, and chemistry. The pool is designed to be comprehensive and diverse in both subject coverage and task types. Unlike generic question-answering (QA) datasets, this scientific data pool captures the fine-grained complexity of scientific tasks, including multi-step reasoning, formula derivation, computational verification, literature retrieval, and complex result integration. It aims to achieve coarse-grained specialization by shifting the training data distribution to scientific domains. In summary, it serves as a large-scale and diverse scientific data engine, providing a solid foundation for subsequent fine-grained problem-answer filtering toward deep capability specialization in science.

### 2.1.2. World-Model-Like Prediction Capability of Strong Models

We leverage the world-model-like prediction capability of strong models to estimate the training value of each scientific problem. Unlike prior work that mainly uses strong models for data quality scoring or sample selection [3, 4], we further exploit their world-model-like prediction capability to assess the training value of candidate problems before generating trajectories. By leveraging the implicit understanding of knowledge boundaries and tool benefits encoded in strong models, we estimate each scientific problem's training value along two dimensions: difficulty and interaction benefit. The difficulty annotation identifies medium-to-hard problems that produce effective training signals, while the interaction-benefit annotation determines whether external objects such as web search, paper retrieval, or code execution can improve performance. This predictive annotation maps raw scientific problems into a fine-grained capability space defined by learning value and tool demand, upgrading problem selection from quality filtering to training-value modeling. This reframes data engineering from simple data collection or hard-problem selection into an explicit process of modeling which problems are worth training on and how they should be learned.

### 2.1.3. Multi-Stage Fine-Grained Filtering Pipeline

We build a multi-stage, fine-grained filtering pipeline to transform scientific data from large-scale pool into high-quality training data. Specifically, we progressively filter scientific problems through quality filtering, difficulty selection, and capability routing. We first remove QA mismatches and ill-formatted questions to ensure that each retained problem is clear, answer-matched, and evaluable. We then leverage the world-model-like prediction capability of strong models to estimate each problem's learning signals, and use the resulting difficulty predictions to select medium-to-hard problems, resulting in around 30K high-quality, challenging, and learnable problems. Finally, based on interaction benefit prediction from strong models, we route each problem to an appropriate reasoning path for training: problems that can be solved primarily through internal knowledge and logical derivation follow a pure-reasoning path, while problems that benefit from web search, literature retrieval, or code execution follow an interaction-reasoning path. Through this multi-stage fine-grained filtering process, data construction moves from coarse scientific-domain coverage to fine-grained training-value modeling, enabling more efficient and effective training with fewer but higher-quality data.

## 2.2. Scientific Interaction Framework

In this section, a deeply specialized scientific interaction framework is constructed for scientific reasoning and research. We first abstract the key operations in scientific reasoning and research into distinct interaction objects, and define an *action-observation contract* between the model and the interaction framework: which objects the model can access, which actions it can perform on these objects, what observations it receives, and how these observations are fed back into the reasoning process. We then obtain high-quality interactive reasoning trajectories through *Chain-of-Interaction* distillation and use them for model training. After training, the model no longer relies solely on its internal capability to complete reasoning, but acquires the ability to actively select interaction objects, analyze external observations, and iteratively update its reasoning state. Furthermore, we find that based on a finite set of interaction objects, scientific interaction patterns can emerge, including *evidence-seeking*, *verification-seeking*, and *hybrid closed-loop* scientific interaction modes. This extends the model from a pure reasoner that primarily relies on parametric knowledge into a scientific agent capable of autonomously acquiring evidence, performing verification, incorporating feedback, and integrating conclusions. The overview of this framework is illustrated in Fig. 4.

### 2.2.1. Deeply Specialized Scientific Interaction Objects

Deeply specialized scientific interaction objects constitute the core of the interaction framework. Scientific reasoning and research are not closed-form text generation processes. Instead, they are interactive processes where researchers continuously search for external knowledge, extract relevant evidence, perform verification, and revise hypotheses based on the observations. Therefore, deep specialization of scientific capability is reflected first in whether the model is equipped with interaction objects that match the required operations, rather than in whether it memorizes more domain knowledge. In this work, we extract a set of atomic capabilities from scientific reasoning and research behaviors, including *source discovery*, *source grounding*, *executable verification*, and *observation integration*. We further instantiate these atomic capabilities as different interaction objects under a unified interaction framework. Specifically, `Web Search` and `Google Scholar` jointly support source discovery. The former provides broad-coverage retrieval over the open web, expanding the model's knowledge boundary, while the latter provides targeted access to papers, problem sources, and academic materials, introducing higher-credibility evidence entry points for scientific questions. `Browser` is responsible for source grounding, enabling the model to obtain compact source-grounded content. `Code Executor`

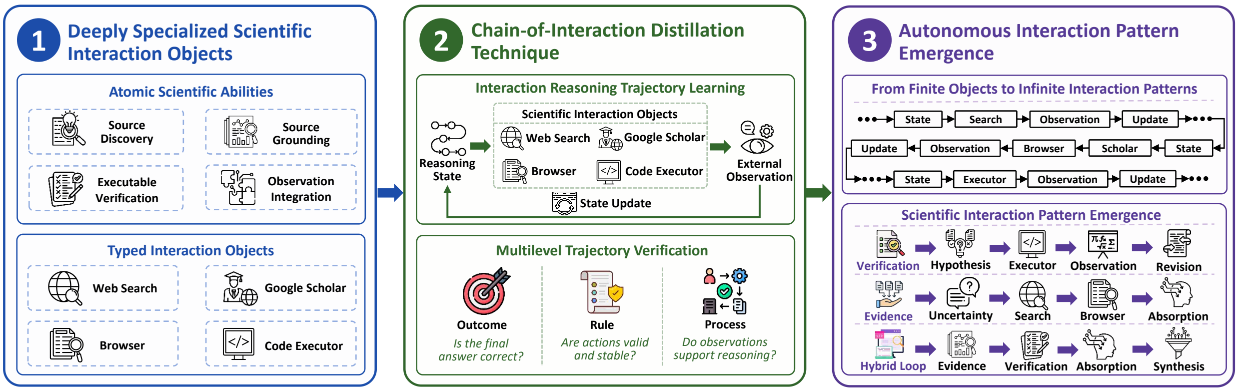


Figure 4 | Overview of the deeply specialized scientific interaction framework. The framework abstracts atomic scientific abilities into typed interaction objects governed by a unified action–observation contract, and uses Chain-of-Interaction distillation to enable high-quality interaction reasoning trajectory learning via multilevel trajectory verification. Built upon this framework, finite interaction objects can realize infinite interaction patterns, and emerge scientific interaction patterns including verification-seeking, evidence-seeking, and hybrid closed-loop interaction patterns.

serves as an executable substrate that supports computation, fitting, simulation, visualization, and multimodal evidence processing.

### 2.2.2. Chain-of-Interaction (CoI) Distillation Technique

Chain-of-Interaction distillation technique provides the concrete path for acquiring interactive reasoning ability. Analogous to Chain-of-Thought improving model capability through task decomposition and iterative reasoning, Chain-of-Interaction (CoI) extends model capability through long-horizon autonomous interaction between the model and different objects. Therefore, interaction trajectories should also be considered besides the reasoning trajectories. Through CoI distillation, we construct a high-quality interaction trajectories for training, enabling the model to access retrievable, browsable, and executable external objects during reasoning, and to integrate autonomously obtained observations as the basis for judgment, computation, and conclusion synthesis. Moreover, because interaction trajectories are verifiable, we ensure trajectory quality at three levels: outcome, rule, and process. At the outcome level, we retain trajectories that lead to correct final answers. At the rule level, we remove unstable samples, such as trajectories containing repeated ineffective tool calls, vacuous responses, or overly short answers. At the process level, we focus on the rationality of interaction-object selection, the information gain provided by the returned observations, and whether the model updates its reasoning state after receiving these observations. In this way, CoI distillation enables the model to form high-quality consistent interactions with external objects and continuously optimize its reasoning capability throughout the interaction process.

### 2.2.3. Autonomous Interaction Pattern Emergence

The abstracted finite set of interaction objects can promote the emergence of autonomous interaction patterns. We find that the interaction patterns in CoI trajectories are neither mechanical executions of fixed workflows nor linear stacks of interaction objects. Instead, they arise from the model's autonomous composition of a finite set of interaction objects during interactive reasoning, forming stable and reusable scientific interaction patterns. Specifically, the model selects, connects, and recombines different interaction objects according to its current uncertainty, resulting in the interaction trajectory gradually exhibiting a stable structure for scientific problem solving. For example, in *verification-*

*seeking* trajectories, the model transforms intermediate hypotheses into executable feedback and advances the solution through sandbox execution, numerical observations, and reasoning revision. In *evidence-seeking* trajectories, the model reduces source-level or knowledge-level uncertainty through retrieval, browsing, and evidence analysis. In more complex research-oriented trajectories, the model forms a *hybrid closed-loop* pattern, coupling external evidence acquisition and executable verification within the same reasoning path. These findings shift the analytical focus from interaction type, interaction object count, or number of interaction turns to the quality of interaction organization. They demonstrate that the key to interaction intelligence is not how many times an interaction object is invoked, but whether the interaction produces useful observations, and whether the interaction objects are organized into an efficient, traceable, and absorbable closed loop.

### 2.3. Systematic Training Strategy

In this section, we aim to develop a systematic training strategy that transforms a general-purpose Qwen3.5-35B-A3B [5] model into an agentic reasoner that can interact with the deeply specialized scientific interaction framework. Thus, we introduce two mechanisms: a *Training-Inference Consistency Constraint* and *Hybrid Progressive Chain-of-Interaction (CoI) Training*. The former guarantees a unified interaction framework during both training and inference, mitigating capability degradation caused by shifts in the interaction interface and enabling the trained model to fully leverage the deeply specialized scientific interaction framework. The latter jointly trains on reasoning and interaction trajectories, allowing the model to acquire both intrinsic reasoning and extrinsic interaction capability. By progressively introducing high-quality interaction trajectories, the model is further supervised to internalize effective interaction patterns. Fig. 5 provides an overview of our systematic training paradigm. Different from the conventional scaling paradigm that relies primarily on increasing model size, data volume, and training computation, this strategy jointly considers LLMs, interaction objects, and autonomous interaction processes. As a result, it can remarkably surpass frontier models in scientific reasoning and research tasks with less data, small model, and Low-cost training.

#### 2.3.1. Training-Inference Consistency Constraint

In systematic training strategy, consistency between training and inference is a crucial prerequisite for strong interactive reasoning capability. In this work, the model is intended to operate within a deeply specialized interaction framework for scientific tasks. This framework defines the model's interactive state space and behavioral boundary through multiple components, such as *prompt*, *interaction objects*, and *observation*. Consequently, the model's inference-time performance depends critically on the match between the inference environment and the interaction structure during training. Our experiments in Sec. 4.1 show that, when the inference-time prompt or interaction objects deviate from the training-time setup, the model suffers from a significant performance degradation. In contrast, the model achieves optimal performance when the inference-time configuration is fully aligned with the training-time configuration. This observation indicates that strict consistency of the interaction structure across training and inference is essential for fully exploiting the potential of a deeply specialized interaction framework. Therefore, we adopt the same interaction framework consistently across all training stages and inference, maximizing the performance.

#### 2.3.2. Hybrid Progressive CoI Training

Building on training-inference consistency, we further design a hybrid progressive Chain-of-Interaction (CoI) training strategy to strengthen both reasoning and interaction capabilities. During training, reasoning and interaction trajectories are adopted as supervision. Both types of trajectories are

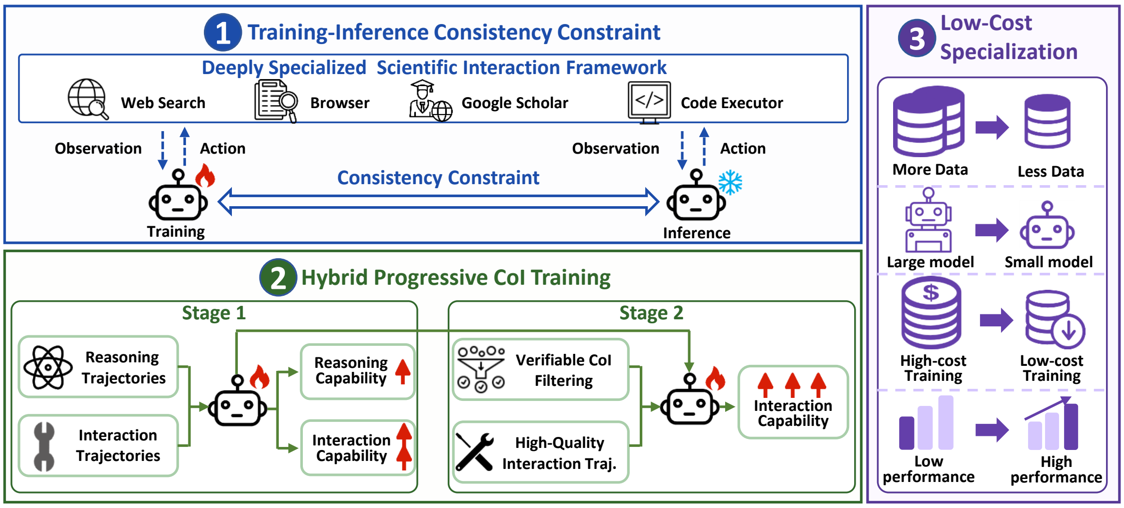


Figure 5 | Overview of the systematic training paradigm. The paradigm introduces training-inference consistency constraints and adopts a two-stage hybrid progressive CoI training strategy, enabling strong scientific reasoning and research capabilities under a low-cost condition of less data, small model, and low-cost training.

optimized using a unified supervision fine-tuning objective:

$$\mathcal{L}_{\text{SFT}} = -\sum_{\tau \in \mathcal{D}} \sum_{t \in \mathcal{M}(\tau)} \log p_\theta(y_t \mid y_{<t}),$$

where $y_t$ denotes the target output token at time step $t$, $\mathcal{D}$ denotes the training data, which may consist of reasoning trajectories, interaction trajectories, or a mixture of both, and $\mathcal{M}(\tau)$ denotes the assistant-token mask for loss computation. For reasoning trajectories, the loss is computed over reasoning steps and final answers. For interaction trajectories, it is additionally computed over tool invocation tokens. User queries, system prompts, tool descriptions, and tool-returned observations are used only as contextual inputs and are excluded from loss computation.

Based on this unified SFT objective, we adopt a two-stage training to progressively build the model's reasoning and interaction capabilities. In the first stage, we mix reasoning trajectories with interaction trajectories, enabling the model to acquire both stable intrinsic scientific reasoning and basic interaction capability. Specifically, following the data filtering strategy described in Sec. 2.1, we select training samples from medium-to-hard scientific problems. We further categorize the problems and their solution trajectories according to the interaction-benefit, yielding a mixture of correct reasoning and interaction trajectories, and use them for the first-stage SFT. In the second stage, we further enhance the model's interaction reasoning capability by selecting higher-quality interaction trajectories. In detail, we retain trajectories with gap-driven interaction-object invocations, observations used in subsequent reasoning, and an efficient, traceable, and absorbable closed loop. We then conduct SFT on this subset, further internalizing interaction reasoning behaviors within the deeply specialized scientific interaction framework. Implementation details for the two-stage training procedure are provided in Appendix A.

### 2.3.3. Low-Cost Deep Specialization

A key property of our training paradigm is its cost efficiency. Under a relatively small model and less data, significant performance improvements can be achieved only using SFT. Unlike conventional general-purpose LLM training, where the model itself is responsible for memorizing broad open-domain knowledge and adapting diverse environments, our paradigm shifts the majority of the optimization burden to the whole agentic system, consisting of the LLM, interaction objects, and the autonomous interaction process. By deeply specializing the interaction space, the model is relieved from acquiring fully general capabilities in an open-ended environment. Instead, training focuses strictly on policy convergence and observation utilization within finite interaction objects, stable interaction protocols, and structured scientific task formats. This shift from broad capability expansion to targeted policy convergence yields highly concentrated, information-dense learning signals. Overall, our paradigm replaces the conventional heavy capability scaling path of increasing model size, data volume, and training compute with a low-cost learning path driven by an extended-mind-inspired agentic system. It substantially improves performance on scientific reasoning and research tasks at low cost under a small model, less data, SFT-only training setting.

# 3. Experiments

## 3.1. Evaluation Settings

To comprehensively evaluate the capabilities of our extended-mind-inspired agentic system, we construct a multi-level and multidimensional evaluation framework targeting both scientific intelligence, defined as multidisciplinary scientific reasoning and research capabilities, and general intelligence, represented by a broad range of common general-purpose tasks. Under a unified evaluation protocol, we compare our ExoMind with a diverse set of state-of-the-art open-source and closed-source models.

### 3.1.1. Evaluation Datasets of Scientific Intelligence

To comprehensively evaluate models' scientific intelligence, we assess their capabilities along two complementary dimensions: scientific research and scientific reasoning. Specifically, we evaluate models on a suite of multidisciplinary benchmarks that cover both closed-set, competition-level problem solving and open-ended, research-oriented scientific tasks. As shown in Fig. 6, these benchmarks span diverse scientific domains and collectively provide a broad testbed for measuring models' ability to reason from internal scientific knowledge, acquire external evidence, perform complex argumentation, and carry out interactive verification.

The scientific research benchmarks include HLE [25], FS-R (FrontierScience-Research) [26], CMT (CMT-Benchmark) [27], and CritPt [28]. These benchmarks cover expert-level frontier knowledge, open-ended research sub-tasks, and executable verification challenges. Specifically, HLE targets questions at the frontier of human knowledge; FS-R focuses on PhD-level research sub-tasks; CMT emphasizes specialized modelling and computational problems in condensed-matter theoretical physics; and CritPt further stresses compound research challenges with code-executable verification. These benchmarks more closely reflect real scientific research workflows, enabling a systematic assessment of models' scientific research capabilities in evidence acquisition, complex argumentation, and interactive reasoning.

The scientific reasoning benchmarks include AMO (AMO-Bench) [29], IMO (IMO-AnswerBench) [30], HiPhO [31], and FS-O (FrontierScience-Olympiad) [26]. These benchmarks primarily assess model performance on challenging closed-set, competition-level problems across scientific disciplines. Specifically, AMO and IMO focus on mathematical derivation and exact answer generation; HiPhO is derived from real physics olympiad problems and provides fine-grained process-level scoring; and FS-O covers olympiad-level multidisciplinary science problems. Together, these benchmarks measure scientific reasoning capabilities grounded in endogenous scientific knowledge, long-chain reasoning, formula derivation, and the generation of precise and verifiable answers.

Table 1 | Detailed configurations of evaluated models from different providers. The green background represents the models for the providers are open-source. The symbol ~ indicates that the model size is estimated based on the Incompressible Knowledge Probes (IKP) method proposed in [1].

| Provider | Model | Release | Size | Arch. | Open |
|---|---|---|---|---|---|
| OpenAI | GPT-5.5 (xhigh) [6] | 2026-04 | ~9.7T | - | × |
| | GPT-5.4 (xhigh) [6] | 2026-03 | ~2.2T | - | × |
| Google | Gemini-3.1-Pro-Preview [7] | 2026-02 | ~9.0T | - | × |
| | Gemini-3.5-Flash-Thinking [7] | 2026-05 | ~6.6T | - | × |
| | Gemma-4-31B [8] | 2026-04 | 31B | Dense | ✓ |
| Anthropic | Claude-Opus-4.8 [9] | 2026-05 | ~572B | - | × |
| | Claude-Opus-4.8-Thinking [9] | 2026-05 | ~936B | - | × |
| DeepSeek | DeepSeek-V4-Pro (Max) [10] | 2026-04 | 1.6T / A49B | MoE | ✓ |
| | DeepSeek-V4-Flash (Max) [10] | 2026-04 | 284B / A13B | MoE | ✓ |
| Moonshot AI | Kimi-K2.6 [11] | 2026-04 | 1T / A32B | MoE | ✓ |
| | Kimi-K2.5 [12] | 2026-01 | 1T / A32B | MoE | ✓ |
| Zhipu AI | GLM-5.2 [13] | 2026-06 | 744B / A40B | MoE | ✓ |
| | GLM-5.1 [13] | 2026-04 | 744B / A40B | MoE | ✓ |
| | GLM-5 [13] | 2026-02 | 744B / A40B | MoE | ✓ |
| MiniMax | MiniMax-M3 [14] | 2026-06 | 428B / A23B | MoE | ✓ |
| | MiniMax-M2.7 [15] | 2026-05 | 229B / A10B | MoE | ✓ |
| Qwen | Qwen3.7-Max [16] | 2026-05 | ~685B | - | × |
| | Qwen3.6-Max-Preview [17] | 2026-04 | ~685B | - | × |
| | Qwen3.6-Plus [18] | 2026-04 | ~524B | - | × |
| | Qwen3.6-35B-A3B [19] | 2026-04 | 35B / A3B | MoE | ✓ |
| | Qwen3.5-397B-A17B [20] | 2026-02 | 397B / A17B | MoE | ✓ |
| | Qwen3.5-122B-A10B [5] | 2026-02 | 122B / A10B | MoE | ✓ |
| | Qwen3.5-35B-A3B [5] | 2026-02 | 35B / A3B | MoE | ✓ |
| Shanghai AI Lab | P1-VL-235B-A22B [21] | 2026-02 | 235B / A22B | MoE | ✓ |
| | P1-VL-30B-A3B [21] | 2026-02 | 30B / A3B | MoE | ✓ |
| | P1-235B-A22B [22] | 2025-11 | 235B / A22B | MoE | ✓ |
| | P1-30B-A3B [22] | 2025-11 | 30B / A3B | MoE | ✓ |
| | SU-01 [23] | 2026-05 | 30B / A3B | MoE | ✓ |
| | Intern-S2-Preview [24] | 2026-05 | 35B / A3B | MoE | ✓ |

### 3.1.2. Evaluation Datasets of General Intelligence

To assess whether enhanced scientific intelligence transfers to broader general capabilities, we further evaluate our ExoMind on a diverse suite of general intelligence benchmarks. This evaluation is designed to characterize the impact of deep specialization on general reasoning, question answering, code generation, deep search, tool use, and real-world agentic task execution. As shown in Fig. 7, these benchmarks span a broad range of scenarios and cover diverse domains, including science and engineering, medicine and health, humanities and history, psychology, and other general disciplines.

The general intelligence benchmarks include MMLU-PRO [32], GPQA-Diamond [33], MedMCQA [34], LiveCodeBench [35], $\tau^2$-Bench [36], and xbench [37]. MMLU-PRO, GPQA-Diamond, and MedMCQA primarily evaluate knowledge understanding and question-answering ability across broad, expert-

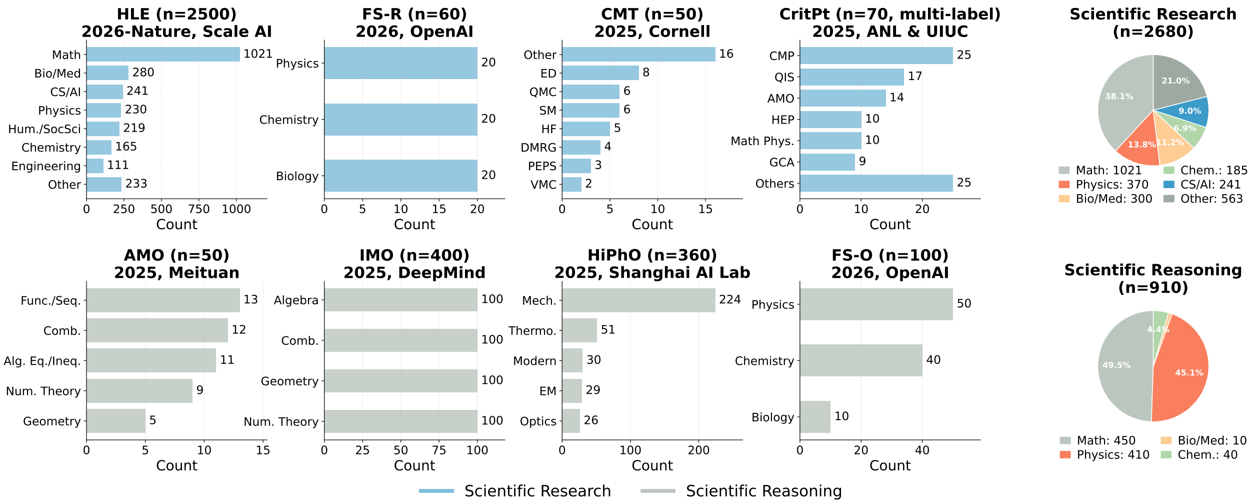


Figure 6 | Dataset distributions for scientific intelligence evaluation. Bar plots represent benchmark-specific domains, subdomains, methods, or topic tags, while pie charts summarize samples into broad scientific reasoning and scientific research domains. For multi-label benchmarks, bar counts correspond to label occurrences and may exceed the number of questions.

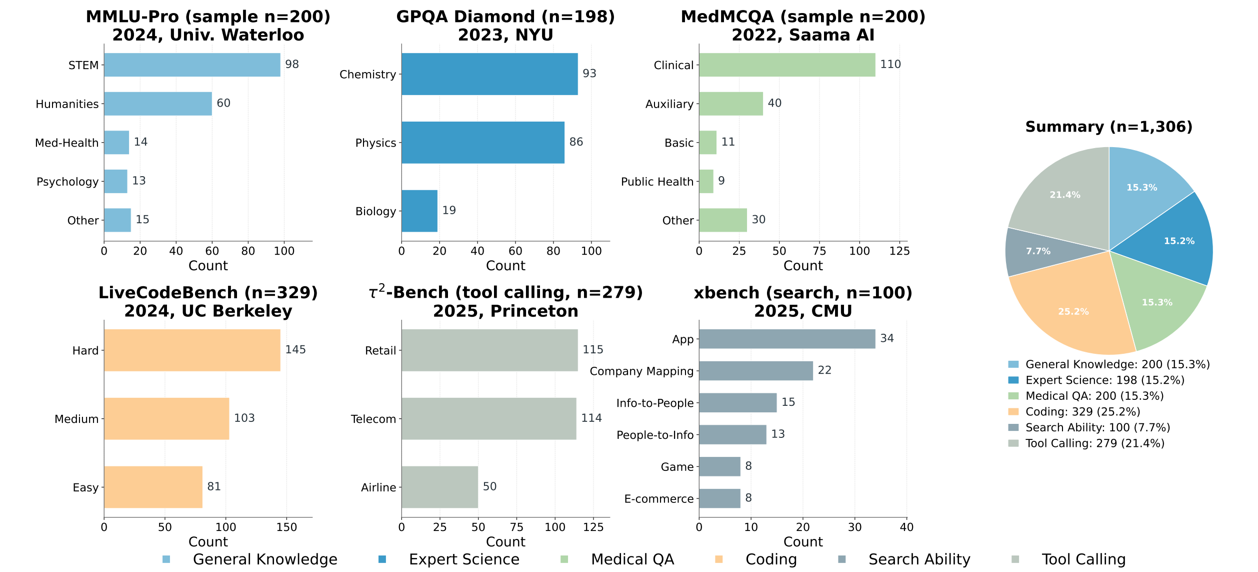


Figure 7 | Dataset distributions for general intelligence evaluation. Panels show benchmark-specific categories, including domains, disciplines, difficulty levels, industries, and task types. The summary pie chart shows the capability distribution across evaluated samples of different benchmarks.

level, and medical domains. LiveCodeBench evaluates code-generation capability through verifiable programming tasks. $\tau^2$-Bench and xbench further assess tool use, multi-turn interaction, deep search, and real-world agent capabilities. Together, these benchmarks provide a multidimensional evaluation of whether scientific specialization preserves and improves general intelligence.

### 3.1.3. Detailed Configurations of Evaluated Models

To comprehensively evaluate the scientific intelligence of our ExoMind, we systematically benchmark representative open-source and closed-source models from different providers, spanning diverse parameter scales and model architectures, as shown in Tab. 1. This comparison provides a broad overview of the capability landscape of current mainstream and frontier LLMs on scientific reasoning and research tasks.

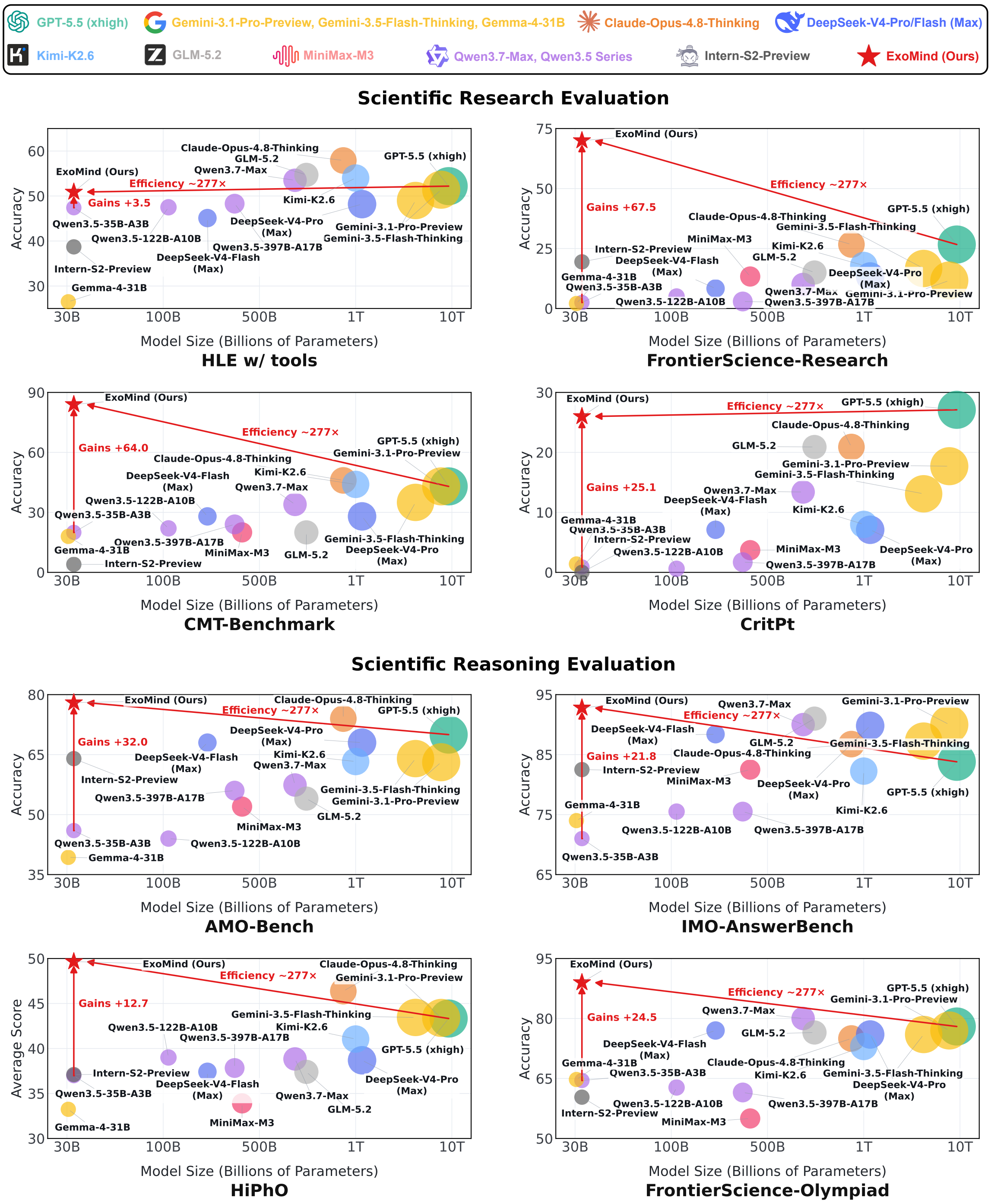


Figure 8 | **Our ExoMind rivals frontier models on scientific research and reasoning benchmarks.** ExoMind consistently improves over the base model on all eight datasets, and surpasses leading open- and closed-source models on most of them. Despite its compact 35B scale, it achieves frontier-level performance with roughly 277× fewer parameters than GPT-5.5 (estimated at ~ 9.7 T parameters by IKP [1]). These results suggest a practical path toward democratizing scientific intelligence through an extended-mind-inspired agentic system.

Table 2 | Main results on scientific intelligence evaluation. Our ExoMind is compared with frontier open-source and closed-source models on various scientific research and scientific reasoning datasets. Higher scores indicate better performance. The green background marks open-source model results. In each column, bold underlined scores denote the best results, and underlined scores denote the second-best results. "–" denotes unavailable results.

| Model | Scientific Research | | | | Scientific Reasoning | | | | Avg. |
|---|---|---|---|---|---|---|---|---|---|
| | HLE w/ tools | FS-R | CMT | CritPt | AMO | IMO | HiPhO | FS-O | |
| GPT-5.5 (xhigh) [6] | 52.2 | 26.7 | 43.0 | **27.1** | 70.0 | 83.8 | 43.3 | 78.0 | 53.0 |
| GPT-5.4 (xhigh) [6] | 52.1 | 33.0 | 42.0 | 23.4 | 66.0 | 85.4 | 42.9 | 78.0 | 52.9 |
| Gemini-3.1-Pro-Preview [7] | 51.4 | 11.7 | 43.0 | 17.7 | 63.1 | 90.0 | 43.4 | 77.0 | 49.7 |
| Gemini-3.5-Flash-Thinking [7] | 49.0 | 16.7 | 35.0 | 13.1 | 64.0 | 87.3 | 43.5 | 76.0 | 48.1 |
| Gemma-4-31B [8] | 26.5 | 2.1 | 18.0 | 1.4 | 39.3 | 74.0 | 33.2 | 64.8 | 32.4 |
| Claude-Opus-4.8 [9] | – | 12.5 | 28.0 | 20.4 | 72.0 | 86.8 | 44.5 | 74.0 | 53.0 |
| Claude-Opus-4.8-Thinking [9] | **57.9** | 26.7 | 46.0 | 20.9 | 74.0 | 86.8 | 46.4 | 75.0 | 54.2 |
| DeepSeek-V4-Pro (Max) [10] | 48.2 | 13.3 | 28.0 | 7.1 | 68.0 | 89.8 | 38.7 | 76.0 | 46.1 |
| DeepSeek-V4-Flash (Max) [10] | 45.1 | 8.3 | 28.0 | 7.1 | 68.0 | 88.4 | 37.4 | 77.0 | 44.9 |
| Kimi-K2.6 [11] | 54.0 | 17.9 | 44.0 | 8.0 | 63.3 | 82.3 | 41.1 | 73.0 | 47.9 |
| Kimi-K2.5 [12] | 50.2 | 13.3 | 20.0 | 3.1 | 61.8 | 81.8 | 39.3 | 71.0 | 42.6 |
| GLM-5.2 [13] | 54.7 | 15.0 | 20.0 | 20.9 | 54.0 | 91.0 | 37.4 | 76.5 | 46.2 |
| GLM-5.1 [13] | 52.3 | 10.0 | 32.0 | 4.6 | 47.4 | 83.8 | 35.9 | 70.0 | 42.0 |
| GLM-5 [13] | 50.4 | 8.8 | 22.0 | 2.0 | 49.0 | 82.5 | 36.3 | 65.8 | 39.6 |
| MiniMax-M3 [14] | – | 13.3 | 20.0 | 3.7 | 52.0 | 82.5 | 33.9 | 55.0 | 37.2 |
| MiniMax-M2.7 [15] | – | 7.1 | 10.0 | 0.6 | 50.0 | 71.8 | 31.3 | 59.5 | 32.9 |
| Qwen3.7-Max [16] | 53.5 | 10.0 | 34.0 | 13.4 | 57.4 | 90.0 | 38.8 | 80.0 | 47.1 |
| Qwen3.6-Max-Preview [17] | 51.0 | 15.0 | 28.0 | 3.7 | 44.0 | 76.8 | 36.5 | 69.3 | 40.5 |
| Qwen3.6-Plus [18] | 50.6 | 5.8 | 26.0 | 2.9 | 48.0 | 75.5 | 40.9 | 66.3 | 39.5 |
| Qwen3.6-35B-A3B [19] | 36.2 | 2.9 | 20.0 | 0.3 | 58.8 | 78.0 | 37.7 | 60.3 | 36.8 |
| Qwen3.5-397B-A17B [20] | 48.3 | 2.9 | 24.0 | 1.7 | 56.0 | 75.5 | 37.8 | 61.5 | 38.5 |
| Qwen3.5-122B-A10B [38] | 47.5 | 5.0 | 22.0 | 0.6 | 44.0 | 75.5 | 39.0 | 62.8 | 37.0 |
| Qwen3.5-35B-A3B [5] | 47.4 | 2.5 | 20.0 | 0.9 | 46.0 | 71.0 | 37.0 | 64.5 | 36.2 |
| P1-VL-235B-A22B [21] | – | 6.7 | 14.0 | 1.4 | 47.5 | 70.6 | 39.3 | 64.3 | 34.8 |
| P1-VL-30B-A3B [21] | – | 4.6 | 10.0 | 0.0 | 44.5 | 65.3 | 35.0 | 52.5 | 30.3 |
| P1-235B-A22B [22] | – | 7.5 | 16.0 | 0.0 | 53.5 | 70.8 | 35.9 | 62.0 | 35.1 |
| P1-30B-A3B [22] | – | 4.2 | 12.0 | 0.0 | 44.3 | 66.2 | 32.5 | 54.8 | 30.6 |
| SU-01 [23] | – | 11.7 | 8.0 | 0.0 | 59.8 | 77.5 | 35.0 | 61.5 | 36.2 |
| Intern-S2-Preview [24] | 38.7 | 19.4 | 4.0 | 0.0 | 64.0 | 82.5 | 37.1 | 60.3 | 38.3 |
| **ExoMind** | 50.9 | **70.0** | **84.0** | 26.0 | **78.0** | **92.8** | **49.7** | **89.0** | **67.5** |

## 3.2. Main Benchmark Results

### 3.2.1. Benchmark Results on Scientific Intelligence

As shown in Tab. 2 and Fig. 8, our ExoMind achieves highly competitive results across both scientific research and scientific reasoning benchmarks, outperforming current frontier models on most core evaluations while remaining competitive on the hardest frontier benchmarks. On scientific reasoning tasks, ExoMind reaches 89.0 on FS-O, substantially surpassing the strongest external baseline, Qwen3.7-Max, which achieves 80.0. It also obtains the best results on AMO, IMO, and HiPhO, with scores of 78.0, 92.8, and 49.7, respectively. On scientific research tasks, ExoMind achieves 70.0 on FS-R, significantly exceeding the best external result of 33.0 from GPT-5.4 (xhigh), and reaches 84.0 on CMT, far above the strongest external baseline, Claude-Opus-4.8-Thinking, which scores 46.0. Meanwhile, on HLE and CritPt, ExoMind obtains 50.9 and 26.0, respectively. Although it does not yet fully surpass the strongest frontier closed-source models Claude-Opus-4.8-Thinking and GPT-5.5 (xhigh) on these two highly challenging benchmarks, it remains highly competitive. Overall, these results suggest that progress in scientific intelligence does not have to rely solely on scaling model size. Despite being built on a relatively small Qwen3.5-35B-A3B model, ExoMind enables strong scientific reasoning through systematic data engineering, reasoning framework, and training strategy with a small set of high-quality data and SFT. This demonstrates the potential of low-cost deep specialization as an effective path toward improving scientific intelligence.

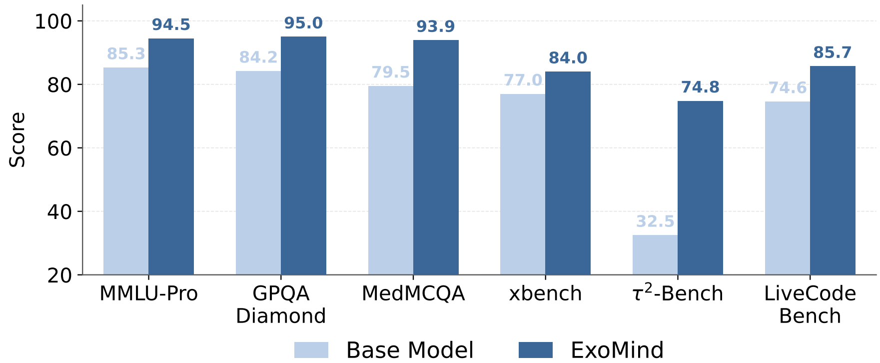


Figure 9 | Results on selected general intelligence benchmarks. We compare the base model (Qwen3.5-35B-A3B) with our ExoMind to illustrate the effect of deep specialization on general capabilities. Higher scores indicate better performance.

### 3.2.2. Benchmark Results on General Intelligence

As shown in Fig. 9, ExoMind consistently improves over the base model across six general intelligence benchmarks. This shows that our model does not trade general capability for scientific specialization; instead, it achieves deep specialization while further improving general intelligence. For general multi-domain reasoning, ExoMind reaches 94.5 on MMLU-PRO. For question answering (QA), it achieves 95.0 on GPQA-Diamond and 93.9 on MedMCQA, covering expert-level scientific QA and medical QA. For code generation, ExoMind improves to 85.7 on LiveCodeBench, indicating that the learned reasoning pattern transfers to verifiable coding tasks. The most pronounced gains appear on interaction-oriented tasks: $\tau^2$-Bench increases from 32.5 to 74.8, and xbench reaches 84.0, showing stronger deep-searching, multi-turn interaction, and real-world task execution ability. Overall, these results indicate that hybrid progressive CoI training does not merely teach task-specific tool invocation. Instead, it induces a broader interactive reasoning capability, where the model learns to condition its actions on the evolving problem state, incorporate external observations, and use them to guide subsequent reasoning. This shared reasoning pattern helps explain why the model can simultaneously improve deep specialization and general-purpose task performance.

## 4. Discussions

### 4.1. Impact of Training-Inference Consistency

As shown in Tab. 3 and Tab. 4, training-inference consistency is a key factor for unlocking scientific interaction reasoning capability. We evaluate this effect from two perspectives: the interaction object configurations and the system prompt configurations. First, in terms of interaction object configurations, ExoMind achieves the best performance when the inference-time interaction object configuration is fully aligned with the training-time setup, reaching 89.0, 70.0, 49.7, and 84.0 on FS-O [26], FS-R [26], HiPhO [31], and CMT [27], respectively. When inference is restricted to a single interaction object or a partial interaction object combination, performance consistently degrades, with the most pronounced drop on FS-R, a research-oriented benchmark that relies heavily on multi-step evidence acquisition and object composition. Second, under the same interaction object configuration, using the training-aligned system prompt yields the most stable performance. Removing or replacing the system prompt leads to clear degradation on tasks such as FS-R and CMT, suggesting that the model depends not only on access to interaction objects, but also on a consistent action–observation protocol for organizing interaction object selection, observation absorption, and subsequent reasoning.

Table 3 | Influence of interaction object configurations. We compare the base model (Qwen3.5-35B-A3B) and our ExoMind under different inference-time configurations of interaction objects, including single object, partial object combinations, and the full-object setting. Object abbreviations: WS = Web Search, GS = Google Scholar, Br = Browser, CE = Code Executor. Within each model group, the best results are bold and underlined, and the second-best results are underlined.

| Configuration | Scientific Research | | Scientific Reasoning | |
|---|---|---|---|---|
| | FS-R | CMT | HiPhO | FS-O |
| *Base Model* | | | | |
| + WS | 3.3 | 12.0 | 36.6 | 54.0 |
| + GS | 6.7 | <u>**16.0**</u> | 37.2 | 52.0 |
| + CE | 3.3 | <u>**16.0**</u> | <u>37.3</u> | 50.0 |
| + GS + Br | <u>15.0</u> | 8.0 | 35.6 | <u>57.0</u> |
| + WS + GS + Br | <u>**18.3**</u> | <u>14.0</u> | <u>**38.5**</u> | <u>**61.0**</u> |
| + WS + GS + Br + CE | <u>**18.3**</u> | 10.0 | <u>**38.5**</u> | <u>**61.0**</u> |
| *ExoMind* | | | | |
| + WS | 16.7 | 34.0 | 34.8 | 67.0 |
| + GS | 20.0 | 24.0 | 38.1 | 66.0 |
| + CE | 10.0 | 30.0 | 40.4 | 73.0 |
| + GS + Br | 31.7 | 56.0 | 39.2 | 73.0 |
| + WS + GS + Br | <u>46.7</u> | <u>82.0</u> | <u>43.8</u> | <u>79.0</u> |
| + WS + GS + Br + CE | <u>**70.0**</u> | <u>**84.0**</u> | <u>**49.7**</u> | <u>**89.0**</u> |

Table 4 | Influence of system prompt configurations under the full-object setting. We evaluate ExoMind with (*w/*) different inference-time system prompt configurations. Prompt abbreviations: NP = No Prompt, IP-1/IP-2 = training-inference Inconsistent Prompt 1/2, and CP = training-inference Consistency Prompt. The best results within the model group are bold and underlined, and the second-best results are underlined.

| Prompt Setting | Scientific Research | | Scientific Reasoning | |
|---|---|---|---|---|
| | FS-R | CMT | HiPhO | FS-O |
| *ExoMind* | | | | |
| *w/* NP | 46.7 | <u>**84.0**</u> | 47.3 | 84.0 |
| *w/* IP-1 | 45.0 | 76.0 | <u>47.9</u> | <u>88.0</u> |
| *w/* IP-2 | <u>50.0</u> | <u>78.0</u> | 44.9 | <u>88.0</u> |
| *w/* CP | <u>**70.0**</u> | <u>**84.0**</u> | <u>**49.7**</u> | <u>**89.0**</u> |

Notably, even under partially misaligned settings, ExoMind still substantially outperforms the base model, indicating that interactive scientific reasoning capability has been effectively injected. Taken together, these results empirically verify the importance of training-inference consistency across the system prompt and interaction object configurations and suggest that similar alignment should be maintained for other parts of the interaction interface, including object protocol, observation formats, and execution harnesses.

## 4.2. The Emergence of Scientific Interaction Patterns

As shown in Tab. 3, increasing the number of available interaction objects does not automatically lead to monotonic performance gains. The key factor is whether the model can organize the atomic capabilities provided by different interaction objects into an observation-guided scientific interaction process. For ExoMind, each single interaction object only provides partial functionality: `Web Search`

**Verification-Seeking**

Problem: (*FrontierScience Olympiad #04*, score: 1.0/1.0).
Compute the exterior electric field of a grounded conducting cylinder in a uniform transverse field.

Reasoning

The model starts from $\Phi = -E_0(r - a^2/r)\cos\theta$, satisfying the grounded boundary and far-field condition. It then isolates the fragile step: converting $\Phi$ to cylindrical components, where the $1/r$ factor and the sign of $E_\theta$ are easy to mishandle.

Action (Code Executor)

`Code Executor` symbolically differentiates $\Phi$, computes $E_r = -\partial_r\Phi$ and $E_\theta = -(1/r)\partial_\theta\Phi$, and simplifies both components before finalizing the field.

Observation (Code Executor)

The action returns $E_r = E_0(1 + a^2/r^2)\cos\theta$ and $E_\theta = E_0(a^2 - r^2)\sin\theta/r^2$, equivalent to $-E_0(1 - a^2/r^2)\sin\theta$.

Analysis

The executable observation resolves the sign ambiguity, verifies that the tangential field vanishes on $r = a$, and recovers $E_0\hat{x}$ as $r \to \infty$. The model uses the observation as a local consistency check rather than a replacement for derivation, and absorbs it into the vector field. This makes the tool call a targeted verification step for a known analytic fragility.

**Evidence-Seeking**

Problem: (*CMT #05*, score: 1.0/1.0).
Identify which of $\mathcal{T}$, $C_{2y}$, and $C_{3z}$ survive in a moiré Hamiltonian with unequal layer masses.

Reasoning

The model partially reasons about $\mathcal{T}$, $C_{2y}$, and $C_{3z}$, but the continuum-model symmetry convention remains uncertain. The key unresolved point is whether $C_{2y}$, as a layer-exchanging operation, survives when $m_b \neq m_t$.

Action (Google Scholar, Web Search & Browser)

`Google Scholar`, `Web Search`, and `Browser` locate symmetry discussions, open the relevant source, and ground the standard continuum-model symmetries in external evidence.

Observation (Google Scholar, Web Search & Browser)

The source evidence supports time reversal and $C_{3z}$-type rotational symmetry in the continuum setting, while clarifying that $C_{2y}$ must be checked against layer exchange rather than assumed from recall.

Analysis

The model combines the grounded evidence with the unequal masses, rather than using retrieved text alone. The source supplies a symmetry prior, while the final decision comes from the problem-specific mass asymmetry. It returns: $\mathcal{T}$ and $C_{3z}$ are preserved, but $C_{2y}$ is broken.

**Hybrid Closed-Loop**

Problem: (*FrontierScience Research #19*, score: 10.0/10.0).
Analyze a quantum-shutter construction for blocking one photon and its general $K$-photon extension.

Reasoning

The model must recover the source-grounded shutter construction and verify the proposed generalization. The uncertainty is both factual, involving the pre-/post-selected states, and algebraic, involving whether transmitted branches vanish after post-selection.

Action (Web Search, Browser & Code Executor)

`Web Search` and `Browser` ground the Aharonov–Vaidman construction and state definitions. `Code Executor` then checks the inner product controlling the $K$-photon blocking condition.

Observation (Web Search, Browser & Code Executor)

The source observation fixes the state form. Symbolic execution verifies that the relevant overlap is proportional to $m - D$; hence it vanishes when $m = D$, establishing the required orthogonality.

Analysis

The trajectory closes the loop by turning grounded evidence into a symbolic condition and verifying it before synthesis. It couples evidence acquisition, formalization, execution, and conclusion integration in one path.

Figure 10 | **Compact examples of CoI trajectories.** Each mini case compresses a complete trajectory into reasoning, action, observation, and analysis blocks. The examples illustrate verification-seeking through executable feedback, evidence-seeking through source grounding, and hybrid closed-loop reasoning through coupled evidence acquisition and symbolic verification.

and `Google Scholar` mainly support source discovery, while `Code Executor` supports executable verification. Without `Browser`, however, the model lacks sufficient source grounding and observation integration, which limits performance on research-oriented tasks such as FS-R and CMT.

When `Google Scholar` is combined with `Browser`, ExoMind improves substantially on CMT, reaching 56.0. Further adding `Web Search` improves FS-R and CMT to 46.7 and 82.0, respectively, suggesting that scientific research tasks require not only discovering relevant information, but also grounding external sources, absorbing observations, and feeding them back into subsequent reasoning. With the full object set enabled, the model achieves the best overall performance. This indicates the emergence of scientific interaction patterns: the model does not simply access more objects, but learns to coordinate source discovery, source grounding, executable verification, and observation integration into reusable scientific problem-solving processes.

In contrast, the base model shows limited and unstable gains even when given access to the same interaction objects. For example, under the full interaction object configuration, the base model achieves only 18.3 on FS-R, far below ExoMind's 70.0. On CMT, this configuration even underperforms some single interaction object settings. This suggests that access alone is insufficient for strong interactive reasoning, and that additional objects may introduce extra complexity in object selection

Table 5 | Effectiveness of our Scientific Interaction Framework (SIF) under the full-objects setting. We compare the strongest models from different providers with our ExoMind across four scientific benchmarks. Results are presented for both without (*w/o*) and with (*w/*) our scientific interaction framework. The best results are bold and underlined, and the second-best results are underlined.

| Model | Scientific Research | | Scientific Reasoning | |
|---|---|---|---|---|
| | FS-R | CMT | HiPhO | FS-O |
| *w/o our SIF* | | | | |
| GPT-5.5 (xhigh) | **26.7** | 43.0 | 43.3 | 78.0 |
| Gemini-3.1-Pro-Preview | 11.7 | 43.0 | 43.4 | 77.0 |
| Claude-Opus-4.8-Thinking | **26.7** | **46.0** | **46.4** | 75.0 |
| DeepSeek-V4-Pro (Max) | 13.3 | 28.0 | 38.7 | 76.0 |
| Kimi-K2.6 | 17.9 | 44.0 | 41.1 | 73.0 |
| GLM-5.2 | 15.0 | 20.0 | 37.4 | 76.5 |
| MiniMax-M3 | 13.3 | 20.0 | 33.9 | 55.0 |
| Qwen3.7-Max | 10.0 | 34.0 | 38.8 | **80.0** |
| Intern-S2-Preview | 19.4 | 4.0 | 37.1 | 60.3 |
| Qwen3.5-35B-A3B | 2.5 | 20.0 | 37.0 | 64.5 |
| *w/ our SIF* | | | | |
| GPT-5.5 (xhigh) | 65.0 | 58.0 | 45.3 | 71.0 |
| Gemini-3.1-Pro-Preview | 31.7 | 70.0 | 43.6 | 77.0 |
| Claude-Opus-4.8-Thinking | 43.3 | 44.0 | 43.5 | 72.0 |
| DeepSeek-V4-Pro (Max) | 43.3 | 60.0 | 45.8 | 86.0 |
| Kimi-K2.6 | 51.7 | 70.0 | 46.5 | 82.0 |
| GLM-5.2 | 58.3 | 48.0 | 44.2 | 74.0 |
| MiniMax-M3 | 46.7 | 28.0 | 39.0 | 73.0 |
| Qwen3.7-Max | 38.3 | 46.0 | 46.1 | 74.0 |
| Intern-S2-Preview | 31.7 | 44.0 | 42.0 | 72.0 |
| Qwen3.5-35B-A3B | 18.3 | 10.0 | 38.5 | 61.0 |
| **ExoMind** | **70.0** | **84.0** | **49.7** | **89.0** |

and interaction organization. After hybrid progressive CoI training, the model can better understand the functional roles of different interaction objects and compose them according to the evolving reasoning state.

To further inspect this emergent organization, we provide compact trajectory-level examples in Fig. 10 (full trajectories are provided in Appendix C). Unlike the quantitative ablation, which measures aggregate effects under different object configurations, these examples expose the internal structure of CoI trajectories: how the model identifies its current uncertainty, selects an appropriate interaction object, absorbs the returned observation, and updates the subsequent reasoning state. The three mini cases correspond to representative scientific interaction patterns. The verification-seeking case uses executable feedback to resolve a sign-sensitive analytic derivation. The evidence-seeking case uses source discovery and source grounding to reduce uncertainty about scientific symmetries before combining the evidence with problem-specific constraints. The hybrid closed-loop case couples external evidence acquisition with symbolic verification, showing how a research-style trajectory can move from source grounding to formal checking and final synthesis.

## 4.3. Effects of the Scientific Interaction Framework

On the scientific research benchmarks, once equipped with this framework, all evaluated models show significant performance improvements, as shown in Tab 5. This substantial growth provides strong evidence that, for complex, open-ended scientific exploration, relying solely on internal parameters and capabilities of a single model is insufficient. It is essential to adopt the new paradigm of intelligence organization proposed in this work: by introducing richer interactive objects and deeper autonomous

Table 6 | Comparison between the base model and our ExoMind across different model scales and interaction-availability settings. We mark the model size for each setting, and report results without (*w/o*) and with (*w/*) inference-time interaction objects (IOs). The best results within each model group are bold and underlined, and the second-best results are underlined.

| Size | Setting | Scientific Research | | Scientific Reasoning | |
|---|---|---|---|---|---|
| | | FS-R | CMT | HiPhO | FS-O |
| *Base Model* | | | | | |
| 9B | *w/o* IOs | 2.5 | <u>16.0</u> | 21.0 | 55.0 |
| 9B | *w/* IOs | <u>10.0</u> | <u>16.0</u> | 30.9 | 45.0 |
| 35B-A3B | *w/o* IOs | 2.5 | <u>**20.0**</u> | <u>37.0</u> | <u>**64.5**</u> |
| 35B-A3B | *w/* IOs | <u>**18.3**</u> | 10.0 | <u>**38.5**</u> | <u>61.0</u> |
| *ExoMind* | | | | | |
| 9B | *w/o* IOs | 7.5 | 18.0 | 39.0 | 67.0 |
| 9B | *w/* IOs | <u>43.3</u> | <u>74.0</u> | <u>43.9</u> | <u>86.0</u> |
| 35B-A3B | *w/o* IOs | 16.7 | 32.0 | 42.7 | 69.0 |
| 35B-A3B | *w/* IOs | <u>**70.0**</u> | <u>**84.0**</u> | <u>**49.7**</u> | <u>**89.0**</u> |

interaction processes, we open up new scaling dimensions that transcend the limits of model size, thereby eliciting emergent capabilities beyond the individual LLMs.

On scientific reasoning benchmarks, the effect is more heterogeneous. As shown in Tab. 5, while most models benefit from the interaction framework, several strong and larger-scale models exhibit performance degradation. For example, Claude-Opus-4.8-Thinking drops from 75.0 to 72.0 on FS-O, and Qwen3.7-Max decreases from 80.0 to 74.0. These results indicate that, for some strong and larger-scale models, many scientific reasoning problems already have compact endogenous solution paths, and additional interaction steps may introduce unnecessary search, verification, or state-update operations. When a model has not been trained to follow the specialized Chain-of-Interaction (CoI) pattern, these interactions may interfere with its native reasoning capability.

This observation complements the training-inference consistency analysis in Sec. 4.1 from a cross-model perspective: while Sec. 4.1 studies ExoMind's dependence on consistent objects and prompts, Tab. 5 shows that frontier models differ in their compatibility with the same scientific interaction framework. Trained directly within the specialized scientific interaction framework with CoI trajectories and train-inference consistency constraints, ExoMind learns when and how to use interaction objects, absorb observations, and return to internal reasoning. Its best performance across both scientific research and reasoning benchmarks indicates that the full benefit of the framework emerges only when the model is aligned with the external interaction space.

### 4.4. CoI Training Improves Standalone Reasoning

As shown in Tab. 6, our hybrid progressive CoI training does not degrade the model's standalone reasoning capability; instead, it leads to consistent positive transfer. Under the standalone evaluation setting, ExoMind improves over the Qwen3.5-35B-A3B base model across all four scientific tasks: FS-O increases from 64.5 to 69.0, FS-R from 2.5 to 16.7, HiPhO from 37.0 to 42.7, and CMT from 20.0 to 32.0. When inference-time interaction objects are available, the performance gains become substantially larger, indicating that external observations further amplify the model's scientific problem-solving capability, while the improvement is not solely attributable to inference-time interaction. Overall, these results suggest that hybrid progressive CoI training not only injects interaction capability, but also partially internalizes the problem decomposition, verification feedback,

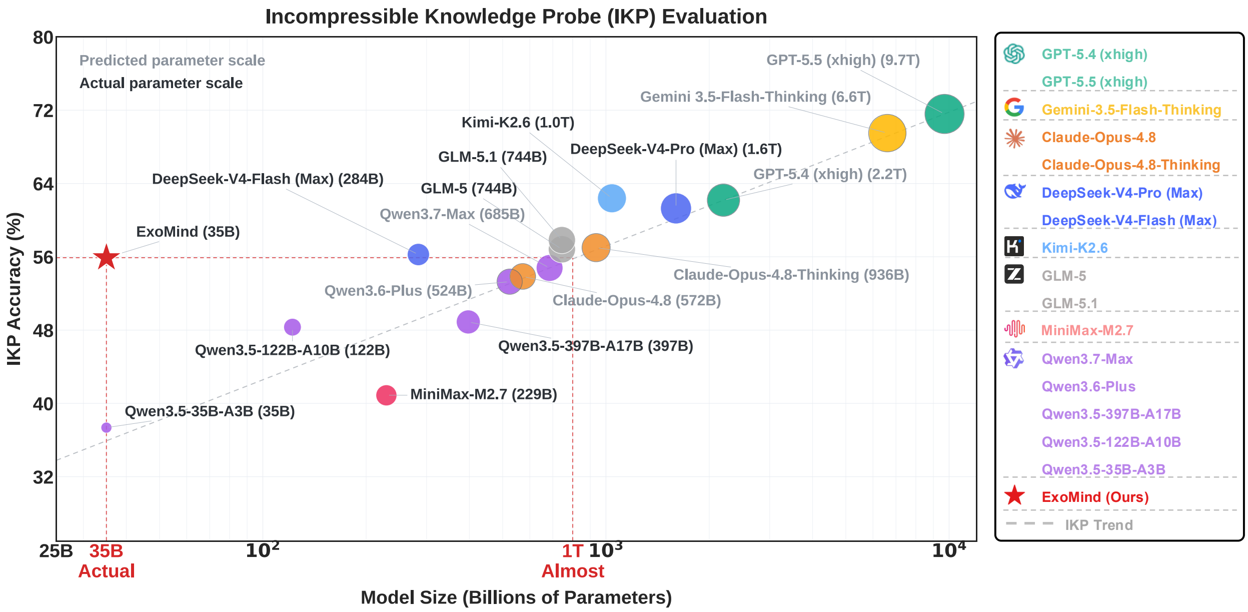


Figure 11 | Incompressible Knowledge Probe (IKP) evaluation of different models. Our ExoMind achieves substantially higher IKP accuracy than its base scale (35B), even reaching performance close to models of around one trillion parameters (1T).

and state-update patterns learned from CoI trajectories into the model's reasoning policy. As a result, the model improves scientific reasoning capability in both standalone and interactive reasoning.

## 4.5. Effectiveness for Smaller Models

As shown in Tab. 6, we further evaluate the applicability of the proposed extended-mind-inspired agentic system to the smaller Qwen3.5-9B model. Under the interactive reasoning setting, the 9B model with our hybrid progressive training achieves substantial improvements over the 9B base model: FS-O increases from 45.0 to 86.0, FS-R from 10.0 to 43.3, HiPhO from 30.9 to 43.9, and CMT from 16.0 to 74.0. The gains are particularly pronounced on FS-R and CMT, which require stronger evidence acquisition, interaction-object composition, and executable verification. These results indicate that the proposed deep-specialization approach does not rely on larger models to be effective. Rather, it can endow smaller models with interactive scientific reasoning capabilities, providing a feasible path toward low-cost specialization in scientific intelligence.

## 4.6. Incompressible Knowledge Probe (IKP) Evaluation

The Incompressible Knowledge Probe (IKP) [1] evaluation demonstrates the amount of incompressible knowledge contained within the model, and the size of the model can be estimated through their elaborated evaluation questions. As shown in Fig. 11, we systematically investigate a range of models using the IKP evaluation. The grey dashed curve captures the empirical relationship between IKP accuracy and model scale, and is used to estimate the IKP-equivalent parameter scale of closed-source models such as GPT-5.5-xhigh [6]. The results show that the proposed agentic model ExoMind substantially improves the IKP accuracy of the base model (35B), reaching a level close to that of models with approximately one trillion parameters (1T). This suggests that long-horizon autonomous interaction with diverse external objects can provide capability gains analogous to those obtained by scaling parameters, data and training. More broadly, these findings indicate that the diversity of external objects and the process of autonomous interaction may constitute a new dimension for scaling intelligence. In other words, we could further improve model capabilities by expanding the number of external objects and strengthening the autonomous interaction process.

# 5. Related Works

## 5.1. Scientific Intelligence

As a broad range of general-purpose benchmarks becomes increasingly saturated and less discriminating for the frontier models, scientific intelligence is emerging as one of the most stringent settings for evaluating advanced intelligence. Unlike conventional evaluations that often emphasize broad knowledge coverage, short-form reasoning, or isolated tool use, scientific intelligence requires systems to integrate deep and heterogeneous domain knowledge, external grounding in tools, and sustained reasoning over long-horizon problem-solving processes. We therefore focus on the evaluation of scientific intelligence and organize existing benchmarks into two complementary types, namely scientific reasoning and scientific research.

Scientific reasoning evaluates difficult but well-specified problems with verifiable answers. In this category, our evaluation considers olympiad-style scientific benchmarks, including AMO-Bench [29], IMO-Bench [30], HiPhO [31], and FrontierScience Olympiad (FS-O) [26]. These benchmarks are difficult for frontier models as they involve not only advanced scientific knowledge, but also frequently requiring non-trivial abstraction, symbolic or quantitative derivation, extended multi-step reasoning, and strict consistency with domain-specific constraints. AMO-Bench [29] consists of original, expert-validated mathematical problems at IMO-level or higher difficulty, with final-answer formats designed for robust automatic grading while reducing contamination from existing competition data. IMO-Bench [30] further evaluates mathematical reasoning through complementary answer, proof and grading tasks. HiPhO [31] focuses on high-level physics reasoning by collecting recent physics olympiad exams, including text-only and diagram-based problems, and aligns evaluation with official marking schemes and human medal thresholds. FS-O [26] extends the olympiad setting to cross-domain scientific reasoning in physics, chemistry, and biology, using original problems written by olympiad medalists and coaches to preserve difficulty, originality, and factual reliability. Together, these benchmarks mark a shift from textbook-style science question answering to competition- and frontier-level scientific reasoning, where performance depends on constructing, maintaining, and verifying coherent reasoning chains across different domains.

Scientific research moves evaluation closer to the frontier of real scientific problems, which probes more open-ended, specialized, and frontier-level tasks under greater uncertainty and process complexity. This category includes Humanity's Last Exam (HLE) [25], FrontierScience Research (FS-R) [26], CMT-Benchmark [27] and CritPt [28]. In details, HLE [25] provides broad expert-level coverage through multimodal, closed-ended questions across many academic disciplines, with answers designed to be unambiguous and automatically gradable while remaining difficult to solve by simple retrieval. FS-R [26] targets PhD-level, open-ended research subtasks written and verified by scientists, and uses rubric-based evaluation to assess intermediate reasoning and problem-solving processes rather than only final answers. CMT-Benchmark [27] focuses on condensed-matter theory at the level of expert researchers, covering topics such as quantum many-body systems, Monte Carlo methods, DMRG, among others, with programmatic grading against expert-supplied ground truth. CritPt [28] further probes unpublished, research-level physics questions spanning multiple modern physics areas, decomposing composite research challenges into checkpoint tasks to test whether models can solve verifiable, leakage-resistant problems rather than reproduce memorized knowledge. Together, these benchmarks characterize a shift from static knowledge recall toward evaluations that approximate authentic scientific practice, emphasizing expert-level reasoning under uncertainty, solving specialized research problems, and producing verifiable intermediate or final outputs.

### 5.2. Agentic AI

As the core technology of artificial intelligence, large language models are undergoing a transition from chat-oriented interaction to deliberative reasoning, and more recently, to agentic capabilities. This transition is supported by the continued scaling of model parameters, training data, training-time and test-time computation [39, 40, 41, 42, 43, 44, 45]. In parallel with these scaling trends, frontier model providers across both proprietary and open-source ecosystems, such as OpenAI, Anthropic, DeepSeek, and Qwen, are increasingly integrating various abilities such as reasoning, reflection, verification, and tool use into unified systems. Early agentic systems achieved such capabilities by coupling base LLMs with carefully designed prompts, workflows, tools, and environment interfaces [46, 47], with representative examples including reasoning–action interleaving [48], autonomous tool acquisition [49], curriculum-driven exploration [50], and verbal self-reflection [51]. However, in these systems, much of the agency is still imposed through hand-designed scaffolding. The emerging direction is therefore to internalize these capabilities within the model's learned behavior, making agency increasingly model-native [52, 53, 54, 47, 55, 56].

In addition, agentic capabilities have been developed and evaluated primarily in domains with clear action spaces and executable environments that provide direct and informative feedback, such as software engineering, web navigation and searching, tool calling [57, 58, 59, 60, 61, 62]. These settings provide natural scaffolds for agents, where actions can be specified, intermediate results can be observed, and outcomes can often be verified automatically. In comparison, the agentic capabilities required for scientific intelligence are structurally different and substantially more demanding. Scientific tasks span heterogeneous disciplines, often lack a single executable environment, and require interaction with specialized objects. Therefore, scientific agentic systems must identify which scientific objects are relevant, decide how to interact with them, interpret intermediate observations, and maintain coherent reasoning over long horizons under uncertainty. Although lots of agents have been proposed for scientific intelligence [63, 64, 65, 66, 67], many remain tied to specific domains, fixed workflows or heavy external orchestration. As a result, generalizable and deeply specialized agentic capability for science remains underexplored.

Our work addresses this research gap by treating scientific intelligence as a primary domain for agentic AI. We introduce an extended-mind-inspired agentic system for scientific reasoning and research by defining atomic scientific capabilities and typed interaction objects, constructing verified Chain-of-Interaction trajectories, and training a model-native interaction policy under strict training-inference consistency. This design shifts scientific specialization from simply increasing model size, data, and training cost or hand-engineering external workflows to equipping the model with a specialized scientific interaction framework. As a result, the abilities of scientific reasoning and research can be learned with limited data, compact models, and short training. This provides a cost-efficient route to agentic scientific intelligence and directly targets the trade-off among customizability, cost efficiency and performance that limits existing approaches.

## 6. Conclusion

In this paper, we propose a new paradigm for organizing intelligence based on the extended-mind thesis. Rather than relying solely on the scaling of a single LLM, our paradigm organizes intelligence through the combination of LLMs, interactive objects, and autonomous interaction processes, enabling scientific capabilities to be specialized through external objects and long-horizon interactions. Further, we introduce **ExoMind** as the first extended-mind-inspired agentic system for scientific intelligence, which includes systematic data engineering, scientific interaction framework, and systematic training strategies. **ExoMind** achieves strong and consistent performance across challenging scientific reasoning and research benchmarks, reaching leading results among frontier open-source and closed-source models while also improving general-purpose capabilities. Notably, these gains are obtained under constrained resource conditions, with less data, small model, and low-cost training.

Looking ahead, the proposed **ExoMind** and extended-mind-inspired paradigm have broad potential for further extension. On the one hand, **ExoMind** can be expanded to cover a more complete scientific workflow, including literature understanding, hypothesis generation, experimental design, process validation, and result analysis. By integrating richer and stronger interactive objects, such as simulation platforms and autonomous laboratories, we can progressively build a highly specialized scientific agentic system capable of handling the fine-grained complexity of real-world scientific applications. On the other hand, this extended-mind-inspired paradigm also holds strong potential for cross-domain extension and can be transferred to other specialized domains to enable expert-level adaptation for specific scenarios with both high performance and low cost. Overall, this work provides a customizable, cost-efficient, and high-performance pathway toward democratizing scientific intelligence and accelerating artificial general intelligence, decoupling advanced intelligence from a small number of large-scale proprietary models and making it broadly accessible and reproducible.

## References

[1] Bojie Li. "Incompressible Knowledge Probes: Estimating Black-Box LLM Parameter Counts via Factual Capacity". In: *arXiv preprint arXiv:2604.24827* (2026).

[2] Andy Clark and David Chalmers. "The extended mind". In: *analysis* 58.1 (1998), pp. 7–19.

[3] Lichang Chen et al. "Alpagasus: Training a better alpaca with fewer data". In: *International Conference on Learning Representations*. Vol. 2024. 2024, pp. 34767–34797.

[4] Mengzhou Xia et al. "LESS: selecting influential data for targeted instruction tuning". In: *Proceedings of the 41st International Conference on Machine Learning*. 2024, pp. 54104–54132.

[5] Qwen Team. *Qwen3.5-35B-A3B Model Card*. https://huggingface.co/Qwen/Qwen3.5-35B-A3B. 2026.

[6] Aaditya Singh et al. "Openai gpt-5 system card". In: *arXiv preprint arXiv:2601.03267* (2025).

[7] Gemini Team et al. "Gemini: a family of highly capable multimodal models". In: *arXiv preprint arXiv:2312.11805* (2023).

[8] Gemma Team et al. "Gemma: Open models based on gemini research and technology". In: *arXiv preprint arXiv:2403.08295* (2024).

[9] Anthropic. *Claude Opus 4.8 System Card*. Accessed: 2026-06-19. 2026.

[10] AI DeepSeek. *Deepseek-v4: Towards highly efficient million-token context intelligence*. 2026.

[11] Moonshot AI. *Kimi K2.6: Advancing Open-Source Coding*. https://www.kimi.com/blog/kimi-k2-6. Accessed: 2026-06-19. 2026.

[12] Kimi Team et al. "Kimi K2. 5: Visual Agentic Intelligence". In: *arXiv preprint arXiv:2602.02276* (2026).

[13] Aohan Zeng et al. "Glm-5: from vibe coding to agentic engineering". In: *arXiv preprint arXiv:2602.15763* (2026).

[14] Xunhao Lai et al. "MiniMax Sparse Attention". In: *arXiv preprint arXiv:2606.13392* (2026).

[15] Aili Chen et al. "The minimax-m2 series: Mini activations unleashing max real-world intelligence". In: *arXiv preprint arXiv:2605.26494* (2026).

[16] Qwen Team. *Qwen3.7*. https://qwen.ai/blog?id=qwen3.7. Accessed: 2026-06-19. 2026.

[17] Qwen Team. *Qwen3.6-Max-Preview*. https://qwen.ai/blog?id=qwen3.6-max-preview. Accessed: 2026-06-19. 2026.

[18] Qwen Team. *Qwen3.6*. https://qwen.ai/blog?id=qwen3.6. Accessed: 2026-06-19. 2026.

[19] Qwen Team. *Qwen3.6-35B-A3B: Agentic Coding Power, Now Open to All*. 2026. URL: https://qwen.ai/blog?id=qwen3.6-35b-a3b.

[20] Qwen Team. *Qwen3.5: Towards Native Multimodal Agents*. 2026. URL: https://qwen.ai/blog?id=qwen3.5.

[21] Yun Luo et al. "P1-VL: bridging visual perception and scientific reasoning in physics Olympiads". In: *arXiv preprint arXiv:2602.09443* (2026).

[22] Jiacheng Chen et al. "P1: Mastering Physics Olympiads with Reinforcement Learning". In: *arXiv preprint arXiv:2511.13612* (2025).

[23] Yafu Li et al. "Achieving gold-medal-level olympiad reasoning via simple and unified scaling". In: *arXiv preprint arXiv:2605.13301* (2026).

[24] Yicheng Zou et al. "Intern-s1-pro: Scientific multimodal foundation model at trillion scale". In: *arXiv preprint arXiv:2603.25040* (2026).

[25] AI Scale, HLE Contributors Consortium, et al. "A benchmark of expert-level academic questions to assess AI capabilities". In: *Nature* 649.8099 (2026), pp. 1139–1146.

[26] Miles Wang et al. "FrontierScience: Evaluating AI's Ability to Perform Expert-Level Scientific Tasks". In: *arXiv preprint arXiv:2601.21165* (2026).

[27] Haining Pan et al. *CMT-Benchmark: A Benchmark for Condensed Matter Theory Built by Expert Researchers*. 2026. arXiv: 2510.05228 [cs.LG]. URL: https://arxiv.org/abs/2510.05228.

[28] Minhui Zhu et al. *Probing the Critical Point (CritPt) of AI Reasoning: a Frontier Physics Research Benchmark*. 2026. arXiv: 2509.26574 [cs.AI]. URL: https://arxiv.org/abs/2509.26574.

[29] Shengnan An et al. *AMO-Bench: Large Language Models Still Struggle in High School Math Competitions*. 2025. arXiv: 2510.26768 [cs.CL]. URL: https://arxiv.org/abs/2510.26768.

[30] Thang Luong et al. *Towards Robust Mathematical Reasoning*. 2025. arXiv: 2511.01846 [cs.CL]. URL: https://arxiv.org/abs/2511.01846.

[31] Fangchen Yu et al. *HiPhO: How Far Are (M)LLMs from Humans in the Latest High School Physics Olympiad Benchmark?* 2025. arXiv: 2509.07894 [cs.AI]. URL: https://arxiv.org/abs/2509.07894.

[32] Yubo Wang et al. "Mmlu-pro: A more robust and challenging multi-task language understanding benchmark". In: *Advances in Neural Information Processing Systems* 37 (2024), pp. 95266–95290.

[33] David Rein et al. "Gpqa: A graduate-level google-proof q&a benchmark". In: *arXiv preprint arXiv:2311.12022* (2023).

[34] Ankit Pal, Logesh Kumar Umapathi, and Malaikannan Sankarasubbu. "Medmcqa: A large-scale multi-subject multi-choice dataset for medical domain question answering". In: *Conference on Health, Inference, and Learning*. PMLR. 2022, pp. 248–260.

[35] Naman Jain et al. "Livecodebench: Holistic and contamination free evaluation of large language models for code". In: *International Conference on Learning Representations*. Vol. 2025. 2025, pp. 58791–58831.

[36] Victor Barres et al. "$\tau^2$-Bench: Evaluating Conversational Agents in a Dual-Control Environment". In: *arXiv preprint arXiv:2506.07982* (2025).

[37] Kaiyuan Chen et al. "xbench: Tracking agents productivity scaling with profession-aligned real-world evaluations". In: *arXiv preprint arXiv:2506.13651* (2025).

[38] Qwen Team. *Qwen3.5: Towards Native Multimodal Agents*. 2026. URL: https://qwen.ai/blog?id=qwen3.5.

[39] Jared Kaplan et al. "Scaling laws for neural language models". In: *arXiv preprint arXiv:2001.08361* (2020).

[40] Jordan Hoffmann et al. "Training compute-optimal large language models". In: *arXiv preprint arXiv:2203.15556* 10 (2022).

[41] Nikhil Sardana et al. "Beyond chinchilla-optimal: Accounting for inference in language model scaling laws". In: *arXiv preprint arXiv:2401.00448* (2023).

[42] Niklas Muennighoff et al. "Scaling data-constrained language models". In: *Advances in Neural Information Processing Systems* 36 (2023), pp. 50358–50376.

[43] Zelin Tan et al. "Scaling behaviors of llm reinforcement learning post-training: An empirical study in mathematical reasoning". In: *arXiv preprint arXiv:2509.25300* (2025).

[44] Runze Liu et al. "Can 1b llm surpass 405b llm? rethinking compute-optimal test-time scaling". In: *arXiv preprint arXiv:2502.06703* (2025).

[45] Shengji Tang et al. "Beyond Gemini-3-Pro: Revisiting LLM Routing and Aggregation at Scale". In: *arXiv preprint arXiv:2601.01330* (2026).

[46] Lei Wang et al. "A survey on large language model based autonomous agents". In: *Frontiers of Computer Science* 18.6 (2024), p. 186345.

[47] Ranjan Sapkota, Konstantinos I Roumeliotis, and Manoj Karkee. "Ai agents vs. agentic ai: A conceptual taxonomy, applications and challenges". In: *Information Fusion* (2025), p. 103599.

[48] Shunyu Yao et al. "React: Synergizing reasoning and acting in language models". In: *arXiv preprint arXiv:2210.03629* (2022).

[49] Timo Schick et al. "Toolformer: Language models can teach themselves to use tools". In: *Advances in Neural Information Processing Systems* 36 (2023), pp. 68539–68551.

[50] Guanzhi Wang et al. "Voyager: An open-ended embodied agent with large language models". In: *arXiv preprint arXiv:2305.16291* (2023).

[51] Noah Shinn et al. "Reflexion: Language agents with verbal reinforcement learning". In: *Advances in Neural Information Processing Systems* 36 (2023), pp. 8634–8652.

[52] Yonadav Shavit et al. "Practices for governing agentic AI systems". In: *Research Paper, OpenAI* (2023).

[53] Alan Chan et al. "Harms from increasingly agentic algorithmic systems". In: *Proceedings of the 2023 ACM Conference on Fairness, Accountability, and Transparency*. 2023, pp. 651–666.

[54] Johannes Schneider. "Generative to agentic ai: Survey, conceptualization, and challenges". In: *arXiv preprint arXiv:2504.18875* (2025).

[55] Mohamad Abou Ali, Fadi Dornaika, and Jinan Charafeddine. "Agentic AI: a comprehensive survey of architectures, applications, and future directions". In: *Artificial Intelligence Review* 59.1 (2025), p. 11.

[56] Jitao Sang et al. "Beyond pipelines: A survey of the paradigm shift toward model-native agentic ai". In: *arXiv preprint arXiv:2510.16720* (2025).

[57] Carlos E Jimenez et al. "Swe-bench: Can language models resolve real-world github issues?" In: *International Conference on Learning Representations*. Vol. 2024. 2024, pp. 54107–54157.

[58] John Yang et al. "Swe-agent: Agent-computer interfaces enable automated software engineering". In: *Advances in Neural Information Processing Systems* 37 (2024), pp. 50528–50652.

[59] Xiao Liu et al. "Agentbench: Evaluating llms as agents". In: *International Conference on Learning Representations*. Vol. 2024. 2024, pp. 52989–53046.

[60] Shuyan Zhou et al. "Webarena: A realistic web environment for building autonomous agents". In: *International Conference on Learning Representations*. Vol. 2024. 2024, pp. 15585–15606.

[61] Grégoire Mialon et al. "Gaia: a benchmark for general ai assistants". In: *International Conference on Learning Representations*. Vol. 2024. 2024, pp. 9025–9049.

[62] Yujia Qin et al. "Toolllm: Facilitating large language models to master 16000+ real-world apis". In: *International Conference on Learning Representations*. Vol. 2024. 2024, pp. 9695–9717.

[63] Qian Huang et al. "MLAgentBench: Evaluating Language Agents on Machine Learning Experimentation". In: *Proceedings of the 41st International Conference on Machine Learning*. Ed. by Ruslan Salakhutdinov et al. Vol. 235. Proceedings of Machine Learning Research. PMLR, 2024, pp. 20271–20309. URL: https://proceedings.mlr.press/v235/huang24y.html.

[64] Ziru Chen et al. "Scienceagentbench: Toward rigorous assessment of language agents for data-driven scientific discovery". In: *International Conference on Learning Representations*. Vol. 2025. 2025, pp. 96934–96990.

[65] Bodhisattwa Prasad Majumder et al. "DiscoveryBench: Towards Data-Driven Discovery with Large Language Models". In: (2024). DOI: 10.48550/arXiv.2407.01725. arXiv: 2407.01725 [cs.CL].

[66] Liqiang Jing et al. "DSBench: How Far Are Data Science Agents to Becoming Data Science Experts?" In: (2024). arXiv: 2409.07703 [cs.AI].

[67] Jiaqi Wei et al. "From ai for science to agentic science: A survey on autonomous scientific discovery". In: *arXiv preprint arXiv:2508.14111* (2025).

[68] Zilin Zhu et al. *slime: An LLM Post-Training Framework for RL Scaling*. https://github.com/THUDM/slime. GitHub repository. Corresponding author: Xin Lv. 2025.

Table 7 | Training hyperparameters for the two-stage Hybrid Progressive CoI Training.

| **Hyperparameter** | **Stage 1** | **Stage 2** |
|---|---|---|
| batch size | 128 | 16 |
| epochs | 5 | 6 |
| Learning rate | $2.0 \times 10^{-5}$ | $1.0 \times 10^{-5}$ |
| Minimum learning rate | $1.0 \times 10^{-6}$ | |
| Optimizer | AdamW | |
| Weight decay | 0.1 | |
| Learning-rate scheduler | Cosine | |
| Warmup ratio | 0.1 | |
| Maximum context length | 65,536 | |
| Training precision | bfloat16 | |

Table 8 | Detailed results on scientific reasoning benchmarks. We report performance on AMO, IMO, HiPhO, and FS-O. For HiPhO, we include scores on individual physics olympiad subsets together with the average and normalized scores. For FS-O, we report subject-level accuracy on biology, chemistry, and physics, as well as the overall accuracy. Higher values indicate better performance.

| Model | AMO | IMO | HiPhO | | | | | | | | | | | | | | | FS-O | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Score | Score | IPhO25 | IPhO24 | APhO25 | Eu25 | Eu24 | NB25 | NB24 | Pan25 | Pan24 | Mech25 | Mech24 | FMA25 | FMA24 | Avg | Norm | Bio | Chem | Phys | Acc |
| GPT-5.5 (xhigh) [6] | 52.4 | 83.8 | 24.0 | 25.9 | 27.3 | 19.0 | 14.8 | 35.0 | 36.7 | 65.1 | 89.5 | **93.0** | 84.0 | 24.0 | **25.0** | 43.3 | 76.7 | 30.0 | 95.0 | 74.0 | 78.0 |
| GPT-5.4 (xhigh) [6] | 66.0 | 85.4 | 27.1 | 26.6 | 28.5 | 4.0 | 25.8 | 36.6 | 35.4 | 73.1 | 81.0 | 89.0 | 83.0 | 24.0 | 24.0 | 42.9 | 76.3 | 60.0 | 80.0 | 80.0 | 78.0 |
| Gemini-3.1-Pro-Preview [7] | 63.1 | 90.0 | 27.8 | 26.8 | 27.0 | 26.0 | 26.0 | 33.6 | 41.2 | 67.8 | 89.0 | 73.0 | 76.0 | **25.0** | **25.0** | 43.4 | 81.2 | 40.0 | 87.5 | 76.0 | 77.0 |
| Gemini-3.5-Flash-Thinking [7] | 64.0 | 87.3 | 28.5 | 26.0 | 27.5 | 24.9 | 26.0 | 35.9 | 38.5 | 65.6 | 88.0 | 72.0 | 83.0 | 24.0 | **25.0** | 43.5 | 80.8 | 30.0 | 81.2 | 81.0 | 76.0 |
| Gemma-4-31B [8] | 39.3 | 74.0 | 19.7 | 24.1 | 24.3 | 6.4 | 19.9 | 28.9 | 23.8 | 40.3 | 52.9 | 69.3 | 78.9 | 22.0 | 21.8 | 33.2 | 61.8 | 25.0 | 66.9 | 71.0 | 64.8 |
| Claude-Opus-4.8 [9] | 72.0 | 86.8 | 25.6 | 25.4 | 27.9 | 16.9 | 22.6 | 35.4 | 37.1 | 74.0 | 90.5 | 85.5 | 91.0 | 24.0 | 23.0 | 44.5 | 78.8 | 40.0 | 77.5 | 78.0 | 74.0 |
| Claude-Opus-4.8-Thinking [9] | 74.0 | 86.8 | 28.7 | 26.8 | **29.7** | 16.2 | 22.8 | 42.0 | 41.5 | 71.1 | 92.0 | **93.0** | 92.0 | 24.0 | 23.0 | 46.4 | 82.0 | 40.0 | 82.5 | 76.0 | 75.0 |
| DeepSeek-V4-Pro (Max) [10] | 68.0 | 89.8 | 21.2 | 23.8 | 27.0 | 13.3 | 21.9 | 36.5 | 33.4 | 60.3 | 68.5 | 76.0 | 78.0 | 21.0 | 22.0 | 38.7 | 70.0 | 30.0 | 85.0 | 78.0 | 76.0 |
| DeepSeek-V4-Flash (Max) [10] | 68.0 | 88.4 | 24.6 | 25.0 | 26.4 | 16.0 | 20.8 | 34.1 | 28.2 | 57.2 | 62.0 | 71.0 | 80.0 | 21.5 | 19.5 | 37.4 | 69.0 | 50.0 | 85.0 | 76.0 | 77.0 |
| Kimi-K2.6 [11] | 63.3 | 82.3 | 24.7 | 25.2 | 27.8 | 15.3 | 23.0 | 33.6 | 36.1 | 52.5 | 86.5 | 77.0 | 84.0 | 24.0 | 24.0 | 41.1 | 75.0 | 30.0 | 85.0 | 72.0 | 73.0 |
| Kimi-K2.5 [12] | 61.8 | 81.8 | 24.1 | 25.2 | 28.2 | 9.3 | 21.1 | 33.2 | 30.7 | 57.5 | 80.4 | 69.5 | 86.4 | 22.9 | 21.6 | 39.3 | 70.8 | 20.0 | 80.0 | 74.0 | 71.0 |
| GLM-5.2 [13] | 54.0 | 91.0 | 21.5 | 25.9 | 26.7 | 15.3 | 20.3 | 34.4 | 35.7 | 58.4 | 62.0 | 63.0 | 83.0 | 20.0 | 20.0 | 37.4 | 68.4 | 40.0 | 87.5 | 75.0 | 76.5 |
| GLM-5.1 [13] | 47.4 | 83.8 | 22.4 | 25.4 | 26.9 | 6.0 | 19.5 | 29.9 | 31.7 | 55.2 | 64.5 | 73.0 | 71.0 | 21.0 | 20.0 | 35.9 | 65.2 | 10.0 | 80.0 | 74.0 | 70.0 |
| GLM-5 [13] | 49.0 | 82.5 | 23.6 | 24.6 | 26.2 | 7.3 | 15.7 | 31.2 | 38.2 | 49.7 | 46.0 | 82.0 | 88.0 | 21.0 | 18.0 | 36.3 | 64.8 | 17.5 | 73.8 | 69.0 | 65.8 |
| MiniMax-M3 [14] | 52.0 | 82.5 | 14.3 | 23.1 | 26.9 | 1.6 | 12.6 | 27.4 | 29.3 | 55.3 | 64.0 | 78.5 | 78.0 | 15.5 | 14.0 | 33.9 | 56.5 | 35.0 | 71.2 | 46.0 | 55.0 |
| MiniMax-M2.7 [15] | – | 71.8 | 18.3 | 20.5 | 21.8 | 4.8 | 13.8 | 24.0 | 27.4 | 48.5 | 53.0 | 60.0 | 82.5 | 18.0 | 14.0 | 31.3 | 54.5 | 30.0 | 71.2 | 56.0 | 59.5 |
| Qwen3.7-Max [16] | 57.4 | 90.0 | 25.6 | 27.6 | 25.9 | 10.9 | 20.1 | 34.7 | 29.4 | 55.7 | 66.0 | 76.0 | 90.0 | 21.0 | 22.0 | 38.8 | 70.5 | 50.0 | 92.5 | 76.0 | 80.0 |
| Qwen3.6-Max-Preview [17] | 44.0 | 76.8 | 22.6 | 26.2 | 27.7 | 8.4 | 19.7 | 35.5 | 30.4 | 52.3 | 57.8 | 70.8 | 84.0 | 20.5 | 19.0 | 36.5 | 66.4 | 22.5 | 76.2 | 73.0 | 69.3 |
| Qwen3.6-Plus [18] | 48.0 | 75.5 | 23.9 | 25.2 | 27.5 | 11.4 | 20.3 | 33.9 | 32.8 | 61.5 | 77.6 | 81.4 | 87.9 | 23.8 | 24.4 | 40.9 | 73.5 | 22.5 | 76.2 | 67.0 | 66.3 |
| Qwen3.6-35B-A3B [19] | 58.8 | 78.0 | 21.8 | 24.4 | 26.1 | 5.4 | 19.1 | 31.0 | 29.3 | 53.2 | 72.0 | 80.1 | 82.4 | 23.1 | 22.8 | 37.7 | 67.5 | 20.0 | 66.2 | 63.5 | 60.3 |
| Qwen3.5-397B-A17B [20] | 56.0 | 75.5 | 24.0 | 25.3 | 27.3 | 11.1 | 20.9 | 33.2 | 33.1 | 57.6 | 75.0 | 68.2 | 70.1 | 23.0 | 22.8 | 37.8 | 69.9 | 7.5 | 71.2 | 64.5 | 61.5 |
| Qwen3.5-122B-A10B [5] | 44.0 | 75.5 | 22.5 | 25.4 | 26.6 | 10.4 | 19.3 | 31.8 | 31.4 | 59.8 | 77.4 | 72.6 | 84.0 | 22.5 | 23.4 | 39.0 | 70.2 | 10.0 | 71.9 | 66.0 | 62.8 |
| Qwen3.5-35B-A3B [5] | 46.0 | 71.0 | 21.5 | 24.5 | 26.4 | 8.7 | 19.7 | 30.6 | 29.4 | 50.4 | 72.9 | 73.2 | 77.4 | 23.0 | 22.9 | 37.0 | 67.5 | 22.5 | 77.5 | 62.5 | 64.5 |
| P1-VL-235B-A22B [21] | 47.5 | 70.6 | 21.0 | 25.3 | 26.9 | 10.7 | 21.5 | 32.5 | 31.9 | 61.7 | 79.8 | 73.3 | 82.0 | 22.5 | 21.9 | 39.3 | 70.4 | 30.0 | 71.2 | 65.5 | 64.3 |
| P1-VL-30B-A3B [21] | 44.5 | 65.3 | 17.9 | 22.5 | 24.6 | 10.6 | 19.0 | 30.2 | 24.6 | 50.8 | 58.6 | 79.1 | 75.5 | 20.8 | 20.3 | 35.0 | 63.1 | 20.0 | 58.8 | 54.0 | 52.5 |
| P1-235B-A22B [22] | 53.5 | 70.8 | 21.2 | 24.7 | 27.4 | 10.8 | 23.0 | 31.8 | 28.4 | 54.7 | 56.7 | 74.7 | 72.9 | 20.9 | 19.4 | 35.9 | 66.2 | 22.5 | 67.2 | 65.8 | 62.0 |
| P1-30B-A3B [22] | 44.3 | 66.2 | 18.5 | 22.3 | 25.4 | 7.4 | 18.8 | 29.2 | 24.0 | 47.0 | 51.4 | 69.5 | 69.1 | 19.6 | 20.0 | 32.5 | 59.8 | 15.0 | 58.1 | 60.0 | 54.8 |
| SU-01 [23] | 59.8 | 77.5 | 21.4 | 25.3 | 27.0 | 9.2 | 20.0 | 33.2 | 29.5 | 52.7 | 61.2 | 61.5 | 77.5 | 19.0 | 17.5 | 35.0 | 63.8 | 25.0 | 69.4 | 62.5 | 61.5 |
| Intern-S2-Preview [24] | 64.0 | 82.5 | 22.0 | 26.4 | 26.9 | 7.9 | 17.9 | 33.3 | 23.7 | 58.6 | 68.1 | 76.0 | 77.6 | 22.1 | 23.1 | 37.1 | 67.2 | 10.0 | 67.5 | 66.0 | 61.0 |
| **ExoMind** | **78.0** | **92.8** | **29.3** | **29.3** | 27.8 | **29.0** | **28.0** | **43.2** | **49.4** | **87.6** | **97.0** | 89.0 | 88.0 | 24.0 | 24.0 | **49.7** | **89.2** | **80.0** | **97.5** | **84.0** | **89.0** |

# A. Training Details

We adopt a two-stage hybrid progressive CoI training. In the first stage, we initialize the model from Qwen3.5-35B-A3B and train it on a mixture of pure-reasoning and tool-augmented trajectories. In the second stage, we initialize from the checkpoint obtained in the first stage and further refine the model using higher-quality tool-interaction trajectories, with the goal of improving tool-use alignment and interactive scientific reasoning capability.

All experiments are implemented with the Slime framework [68] and use full-parameter SFT. Training is conducted on a server equipped with 8 NVIDIA H200 GPUs, each with 141GB of memory. The main training hyperparameters are summarized in Tab. 7. These configurations are selected based on a preliminary hyperparameter search.

Table 9 | Detailed results on scientific research benchmarks. We report tool-augmented performance on HLE, subject-level and overall accuracy on FS-R, and scores on CMT and CritPt. FS-R results are further broken down into biology, chemistry, and physics to assess research-style scientific problem solving across disciplines. Higher values indicate better performance.

| Model | HLE | FS-R | | | | CMT | CritPt |
|---|---|---|---|---|---|---|---|
| | w/ Tools | Bio | Chem | Phys | Acc | Score | Score |
| GPT-5.5 (xhigh) [6] | 52.2 | 35.0 | 35.0 | 10.0 | 26.7 | 43.0 | **27.1** |
| GPT-5.4 (xhigh) [6] | 52.1 | – | – | – | 33.0 | 42.0 | 23.4 |
| Gemini-3.1-Pro-Preview [7] | 51.4 | 10.0 | 25.0 | 0.0 | 11.7 | 43.0 | 17.7 |
| Gemini-3.5-Flash-Thinking [7] | 49.0 | 10.0 | 35.0 | 5.0 | 16.7 | 35.0 | 13.1 |
| Gemma-4-31B [8] | 26.5 | 1.2 | 5.0 | 0.0 | 2.1 | 18.0 | 1.4 |
| Claude-Opus-4.8 [9] | **57.9** | 10.0 | 15.0 | 12.5 | 12.5 | 28.0 | 20.4 |
| Claude-Opus-4.8-Thinking [9] | **57.9** | 20.0 | 45.0 | 15.0 | 26.7 | 46.0 | 20.4 |
| DeepSeek-V4-Pro (Max) [10] | 48.2 | 5.0 | 35.0 | 0.0 | 13.3 | 28.0 | 7.1 |
| DeepSeek-V4-Flash (Max) [10] | 45.1 | 10.0 | 15.0 | 0.0 | 8.3 | 28.0 | 7.1 |
| Kimi-K2.6 [11] | 54.0 | 12.5 | 32.5 | 8.8 | 17.9 | 44.0 | 8.0 |
| Kimi-K2.5 [12] | 50.2 | 5.0 | 30.0 | 5.0 | 13.3 | 20.0 | 3.1 |
| GLM-5.2 [13] | 54.7 | 17.5 | 25.0 | 2.5 | 15.0 | 20.0 | 20.9 |
| GLM-5.1 [13] | 52.3 | 10.0 | 20.0 | 0.0 | 10.0 | 32.0 | 4.6 |
| GLM-5 [13] | 50.4 | 7.5 | 15.0 | 3.8 | 8.8 | 22.0 | 2.0 |
| MiniMax-M3 [14] | – | 15.0 | 20.0 | 5.0 | 13.3 | 20.0 | 3.7 |
| MiniMax-M2.7 [15] | – | 5.0 | 13.8 | 2.5 | 7.1 | 10.0 | 0.6 |
| Qwen3.7-Max [16] | 53.5 | 10.0 | 15.0 | 5.0 | 10.0 | 34.0 | 13.4 |
| Qwen3.6-Max-Preview [17] | 51.0 | 15.0 | 25.0 | 5.0 | 15.0 | 28.0 | 3.7 |
| Qwen3.6-Plus [18] | 50.6 | 5.0 | 12.5 | 0.0 | 5.8 | 26.0 | 2.9 |
| Qwen3.6-35B-A3B [19] | 36.2 | 3.8 | 5.0 | 0.0 | 2.9 | 20.0 | 0.3 |
| Qwen3.5-397B-A17B [20] | 48.3 | 3.8 | 5.0 | 0.0 | 2.9 | 24.0 | 1.7 |
| Qwen3.5-122B-A10B [5] | 47.5 | 7.5 | 6.2 | 1.2 | 5.0 | 22.0 | 0.6 |
| Qwen3.5-35B-A3B [5] | 47.4 | 2.5 | 5.0 | 0.0 | 2.5 | 20.0 | 0.9 |
| P1-VL-235B-A22B [21] | – | 9.4 | 9.4 | 1.2 | 6.7 | 14.0 | 1.4 |
| P1-VL-30B-A3B [21] | – | 6.2 | 7.5 | 0.0 | 4.6 | 10.0 | 0.0 |
| P1-235B-A22B [22] | – | 11.9 | 8.8 | 1.9 | 7.5 | 16.0 | 0.0 |
| P1-30B-A3B [22] | – | 1.2 | 10.0 | 1.2 | 4.2 | 12.0 | 0.0 |
| SU-01 [23] | – | 15.0 | 10.0 | 10.0 | 11.7 | 8.0 | 0.0 |
| Intern-S2-Preview [24] | – | 0.0 | 5.0 | 0.0 | 1.7 | 4.0 | 0.0 |
| **ExoMind** | 50.9 | **60.0** | **75.0** | **75.0** | **70.0** | **84.0** | 26.0 |

## B. Results Details

Detailed results for the scientific reasoning and scientific research benchmarks are reported separately in Tab. 8 and Tab. 9, respectively, with results across all datasets visualized in Fig. 8. These comparisons highlight that our ExoMind can achieve strong scientific intelligence at a compact scale.

On scientific reasoning benchmarks, FS-O, ExoMind achieves 89.0 overall accuracy, substantially surpassing the strongest external baseline, Qwen3.7-Max, which scores 80.0. The subject-level breakdown shows consistent advantages across all three disciplines: ExoMind obtains 80.0 in biology, 97.5 in chemistry, and 84.0 in physics, exceeding the best external results in each subject and indicating robust transfer across olympiad-level scientific domains. On AMO, ExoMind reaches 78.0, outperforming the strongest closed-source baseline, Claude-Opus-4.8-Thinking by 6.0 points. On IMO, it achieves 92.8, exceeding the best competing models by 2.8 points and demonstrating strong formal mathematical reasoning ability. On HiPhO, ExoMind obtains the best average score of 49.7 and the best normalized score of 89.2. The detailed subset results show especially strong performance on IPhO, EuPhO, NBPhO, and PanPhO subsets, where ExoMind achieves the highest scores on most 2024–2025 splits. Since HiPhO evaluates solutions with structured marking schemes, these results suggest that the model improves not only final-answer correctness but also step-wise physics modeling, derivation, and problem-solving quality.

On scientific research benchmarks, where tool-augmented interaction plays a more central role. On

FS-R, ExoMind achieves 70.0 overall accuracy, significantly exceeding the strongest external result of 33.0 from GPT-5.4 (xhigh). The gains are consistent and large across all subjects: ExoMind scores 60.0 in biology, 75.0 in chemistry, and 75.0 in physics, compared with the best external subject-level results of 35.0, 35.0, and 12.5, respectively. This indicates that the model can more effectively combine external information acquisition, evidence verification, and multi-step reasoning in open research-style scientific tasks. On CMT, ExoMind reaches 84.0, far above the strongest external baseline, Kimi-K2.6, which scores 44.0, highlighting its strength on highly specialized condensed-matter and theoretical physics problems that require both domain knowledge and computational reasoning. On HLE, ExoMind obtains 50.9 with tools. Although it does not surpass the strongest frontier proprietary models, it remains competitive with leading systems on broad expert-level closed-set questions. On CritPt, ExoMind scores 26.0, ranking second overall and remaining close to GPT-5.5 (xhigh), which achieves 27.1, while surpassing other strong models such as GPT-5.4 (xhigh) and Claude-Opus-4.8-Thinking. Overall, the detailed tables show that ExoMind delivers substantial improvements on closed-form scientific reasoning tasks, while its largest gains emerge on open-ended scientific research benchmarks where effective interactive reasoning, evidence grounding, and multi-turn interaction are essential. Notably, these achievements are attained with a relatively small backbone—Qwen3.5-35B-A3B—through deep specialized data engineering, a tailored reasoning framework, and a training strategy that leverages only a small set of high-quality SFT data. This demonstrates the potential of low-cost deep specialization as an effective path toward improving scientific intelligence.

## C. Case Study

In this case study, we analyze three representative classes of CoI trajectories: *verification-seeking*, *evidence-seeking*, and *hybrid closed-loop*. These trajectories correspond, respectively, to executable feedback, external evidence grounding, and the coupled organization of evidence acquisition with symbolic verification. The cases illustrate how the model selects interaction objects in concrete scientific problems, incorporates the resulting observations, and organizes heterogeneous interaction objects into traceable, inspectable, and reusable closed loops for scientific problem solving.

**Case Study: Verification-Seeking Pattern in Chain-of-Interaction**

**Problem: (FrontierScience-Olympiad #04, Score: 1.0/1.0)** Consider a long straight conducting cylinder with a radius of $a$, the central axis of the cylinder as the $z$-axis, and located at $x = y = 0$, placed in a uniform external electric field $\overrightarrow{E_0} = E_0\hat{x}$. Apply cylindrical coordinates (where $\theta$ is measured anticlockwise from the polar axis) to this system where the $x$-axis is aligned to the polar axis ($\theta = 0$). Assume all physical quantities are independent of the $z$ coordinate. If the potential at the conducting cylinder is 0, find the electric field distribution $E(r, \theta)$ outside the conductor, in terms of $(E_0, a, r, \theta, \hat{r}, \hat{\theta})$.

**Ground Truth:** $\boxed{\overrightarrow{E}(r,\theta) = E_0(1 + a^2/r^2)\cos(\theta)\hat{r} - E_0(1 - a^2/r^2)\sin(\theta)\hat{\theta}}$

**Reasoning**

We need to solve a classic electrostatics problem: a long straight conducting cylinder of radius $a$, axis along $z$, placed in a uniform external electric field $E_0\hat{x}$. The cylinder is grounded, so the potential on $r = a$ is 0. The problem is two-dimensional because all physical quantities are independent of $z$.
Far from the cylinder the field must approach the uniform applied field $\mathbf{E}_0 = E_0\hat{x}$. Since $\mathbf{E} = -\nabla\Phi$, the far-field potential is

$$\Phi \xrightarrow[r\to\infty]{} -E_0 x = -E_0 r\cos\theta.$$

For a grounded conducting cylinder, the induced correction must cancel the potential on $r = a$, so the standard exterior potential is

$$\Phi(r,\theta) = -E_0(r - a^2/r)\cos\theta = -E_0 r\cos\theta + \frac{E_0 a^2}{r}\cos\theta.$$

Here the second term is $(E_0 a^2/r)\cos\theta$; the factor $r\cos\theta$ in the first term comes from $x = r\cos\theta$. At $r = a$, $\Phi = -E_0(a-a)\cos\theta = 0$. Good.
Now compute the field from $\mathbf{E} = -\nabla\Phi$. In cylindrical coordinates,

$$\nabla\Phi = \hat{r}\frac{\partial\Phi}{\partial r} + \hat{\theta}\frac{1}{r}\frac{\partial\Phi}{\partial\theta},$$

so the two components of the vector field are

$$E_r = -\frac{\partial\Phi}{\partial r},\ E_\theta = -\frac{1}{r}\frac{\partial\Phi}{\partial\theta}.$$

Differentiating by hand gives

$$E_r = E_0(1 + a^2/r^2)\cos\theta,\ E_\theta = -E_0(1 - a^2/r^2)\sin\theta.$$

The $\theta$-component sign is easy to get wrong because of the extra $1/r$ in the cylindrical gradient. Let's verify both components symbolically.

**Action (Code Executor)**

```
import sympy as sp
r, theta, a, E0 = sp.symbols('r theta a E0', positive=True,
real=True)
Phi = -E0*(r - a**2/r)*sp.cos(theta)

# Compute gradient in cylindrical coordinates
dPhi_dr = sp.diff(Phi, r)
dPhi_dtheta = sp.diff(Phi, theta)

# E = - (dPhi/dr) rhat - (1/r) dPhi/dtheta theta hat
Er = -dPhi_dr
Eth = -(1/r)*dPhi_dtheta
sp.simplify(Er), sp.simplify(Eth)
```

**Observation (Code Executor)**

```
E0*(a**2 + r**2)*cos(theta)/r**2, E0*(a**2 - r**2)*sin(theta)/r**2
```

**Reasoning**

The Python output gives:

$$E_r = E_0(a^2 + r^2)\cos(\theta)/r^2 = E_0(1 + a^2/r^2)\cos\theta.$$

$$E_\theta = E_0(a^2 - r^2)\sin(\theta)/r^2 = -E_0(1 - a^2/r^2)\sin\theta.$$

This verifies the manual derivative. In particular, the Python value for $E_\theta$ is $E_0(a^2 - r^2)\sin(\theta)/r^2$. Rewriting $a^2 - r^2 = -(r^2 - a^2)$, we get

$$E_\theta = -\frac{E_0(r^2 - a^2)}{r^2}\sin\theta = -E_0(1 - a^2/r^2)\sin\theta.$$

Thus the negative sign in the $\hat{\theta}$ direction is not a convention mismatch; it follows directly from $E_\theta = -(1/r)\partial_\theta\Phi$.
Collecting the verified components,

$$\mathbf{E}(r,\theta) = E_r\hat{r} + E_\theta\hat{\theta} = E_0\left[\left(1 + \frac{a^2}{r^2}\right)\cos\theta\,\hat{r} - \left(1 - \frac{a^2}{r^2}\right)\sin\theta\,\hat{\theta}\right].$$

Checks: at $r = a$, $E_\theta = 0$, so the conducting surface has no tangential field; as $r \to \infty$, $\mathbf{E} \to E_0(\cos\theta\,\hat{r} - \sin\theta\,\hat{\theta}) = E_0\hat{x}$, recovering the applied uniform field.
Final Answer:

$$\boxed{\mathbf{E}(r,\theta) = E_0\left[\left(1 + \frac{a^2}{r^2}\right)\cos\theta\,\hat{r} - \left(1 - \frac{a^2}{r^2}\right)\sin\theta\,\hat{\theta}\right],\ r \ge a.}$$

**Analysis**

The interaction is a verification-seeking loop: an analytic derivation creates a sign uncertainty, a symbolic execution checks the cylindrical-gradient components, and the returned expressions are absorbed to fix the final vector field.

**Case Study: Evidence-Seeking Pattern in Chain-of-Interaction**

**Problem: (CMT-Benchmark #05, Score: 1.0/1.0)** Given the following single-particle Hamiltonian in the second quantization formalism as

$$\hat{\mathcal{H}}_0 = \sum_{\tau=\pm} \int d^2\boldsymbol{r} \Psi_\tau^\dagger(\boldsymbol{r}) H_\tau \Psi_\tau(\boldsymbol{r}), \tag{1}$$

$$H_\tau = \begin{pmatrix} -\frac{\hbar^2 \boldsymbol{k}^2}{2m_\mathfrak{b}} + \Delta_{\mathfrak{b},\tau}(\boldsymbol{r}) & \Delta_{\mathrm{T},\tau}(\boldsymbol{r}) \\ \Delta_{\mathrm{T},\tau}^\dagger(\boldsymbol{r}) & -\frac{\hbar^2(\boldsymbol{k}-\tau\boldsymbol{\kappa})^2}{2m_\mathfrak{t}} + \Delta_{\mathfrak{t},\tau}(\boldsymbol{r}) \end{pmatrix}, \tag{2}$$

$$\Psi_+(\boldsymbol{r}) = \begin{pmatrix} \psi_{\mathfrak{b},\uparrow,+}(\boldsymbol{r}) \\ \psi_{\mathfrak{t},\downarrow,+}(\boldsymbol{r}) \end{pmatrix}, \quad \Psi_-(\boldsymbol{r}) = \begin{pmatrix} \psi_{\mathfrak{b},\downarrow,-}(\boldsymbol{r}) \\ \psi_{\mathfrak{t},\uparrow,-}(\boldsymbol{r}) \end{pmatrix}, \tag{3}$$

where $\tau = \pm$ represents $\pm K$ valleys, $\hbar \boldsymbol{k} = -i\hbar \partial_{\boldsymbol{r}}$ is the momentum operator, $\boldsymbol{\kappa} = \frac{4\pi}{3a_M}(1, 0)$ is at a corner of the moire Brillouin zone, and $a_M$ is the moire lattice constant. The spin index of the fermion operators $\Psi_\tau$ is both layer and valley dependent. $(m_\mathfrak{b}, m_\mathfrak{t}) = (0.65, 0.35)m_e$ ($m_e$ is the rest electron mass).
The periodic interlayer potential $\Delta_\mathfrak{b}(\boldsymbol{r})$ is parametrized as

$$\Delta_\mathfrak{b}(\boldsymbol{r}) = 2V_\mathfrak{b} \sum_{j=1,3,5} \cos(\boldsymbol{g}_j \cdot \boldsymbol{r} + \psi_\mathfrak{b}), \tag{4}$$

For the interlayer tunneling at $+K$ valley $\Delta_{\mathrm{T},+}(\boldsymbol{r}) = w(1 + \omega e^{i\boldsymbol{g}_2 \cdot \boldsymbol{r}} + \omega^2 e^{i\boldsymbol{g}_3 \cdot \boldsymbol{r}})$, we obtain the tunneling at $-K$ valley as $\Delta_{\mathrm{T},-}(\boldsymbol{r}) = -w(1 + \omega^{-1} e^{-i\boldsymbol{g}_2 \cdot \boldsymbol{r}} + \omega^{-2} e^{-i\boldsymbol{g}_3 \cdot \boldsymbol{r}})$, where $\omega = e^{i\frac{2\pi}{3}}$, and we assume $w$ takes a real value.
Now consider the following symmetries: 1. Time-reversal

$$\hat{\mathcal{T}} \psi_{l,s,\tau}(\boldsymbol{r}) \hat{\mathcal{T}}^{-1} = -\sum_{s'} \epsilon_{ss'} \psi_{l,s',-\tau}(\boldsymbol{r}), \quad \hat{\mathcal{T}} \psi_{l,s,\tau}^\dagger(\boldsymbol{r}) \hat{\mathcal{T}}^{-1} = -\sum_{s'} \epsilon_{ss'} \psi_{l,s',-\tau}^\dagger(\boldsymbol{r}), \quad \hat{\mathcal{T}} i \hat{\mathcal{T}}^{-1} = -i, \tag{5}$$

2. Layer-inversion $C_{2y}$ symmetry. 3. Rotational $C_{3z}$ symmetry.
What are the correct symmetry manifested in the Hamiltonian $\hat{\mathcal{H}}_0$? Choose from the following options. Make sure to only present the final answer in a single boxed LaTeX environment. E.g. $\boxed{a}$
(a) Time-reversal $\mathcal{T}$ symmetry, $C_{2y}$ symmetry, $C_{3z}$ symmetry. (b) Time-reversal $\mathcal{T}$ symmetry, $C_{2y}$ symmetry, no $C_{3z}$ symmetry. (c) Time-reversal $\mathcal{T}$ symmetry, no $C_{2y}$ symmetry, $C_{3z}$ symmetry. (d) Time-reversal $\mathcal{T}$ symmetry, no $C_{2y}$ symmetry, no $C_{3z}$ symmetry. (e) No time-reversal $\mathcal{T}$ symmetry, $C_{2y}$ symmetry, $C_{3z}$ symmetry. (f) No time-reversal $\mathcal{T}$ symmetry, $C_{2y}$ symmetry, no $C_{3z}$ symmetry. (g) No time-reversal $\mathcal{T}$ symmetry, no $C_{2y}$ symmetry, $C_{3z}$ symmetry. (h) No time-reversal $\mathcal{T}$ symmetry, no $C_{2y}$ symmetry, no $C_{3z}$ symmetry.
Your final solution must be placed in a boxed LaTeX environment. Use only the variables and constants given in the problem; do not define additional constants.
Parameters (if any): $a; b; c; d; e; f; g; h$

Functions (if any): None

**Ground Truth:** $\boxed{c}$

---

Reasoning

We need to decide which of $\mathcal{T}$, $C_{2y}$, and $C_{3z}$ are symmetries of the given moire Hamiltonian. The main uncertainty is how the continuum-model symmetry statements should be applied to the layer-dependent masses and valley-dependent tunneling terms in this problem.

...

$C_{2y}$ **symmetry:**

Layer-inversion $C_{2y}$. In a bilayer, a 180-degree rotation about the y-axis exchanges the two layers. It also sends $x \to -x$, $y \to y$, $z \to -z$. In momentum space, $k_x \to -k_x$, $k_y \to k_y$. It might also act on the valley index? Let's think.

...

$C_{3z}$ **symmetry:**

$C_{3z}$ is a 120-degree rotation about the z-axis. It acts on the coordinates as $(x, y) \to (x', y')$. It also acts on the fields. The problem does not specify the action of $C_{3z}$ on the fields.

...

Let's search for "Bistritzer MacDonald Hamiltonian C3z symmetry" or "continuum model twisted bilayer graphene symmetries".

Action (Google Scholar)

```
query:"Bistritzer MacDonald continuum model twisted bilayer
graphene symmetries C2y C3z"; "twisted bilayer graphene
Hamiltonian time reversal C2y C3z"; "moire Hamiltonian layer
inversion C2y rotational C3z"
```

Observation (Google Scholar)

```
Scholar Results:
...
1.[Twisted bilayer graphene.  III. Interacting Hamiltonian and
exact symmetries]
url:https://journals.aps.org/prb/abstract/10.1103/PhysRevB.103.
205413
publicationInfo:BA Bernevig, ZD Song, N Regnault, B Lian -
Physical Review B, 2021 - APS
Date published:2021
Snippet:"...We derive the explicit Hamiltonian of twisted
bilayer graphene ...of the Hamiltonian.  First, we show that all
projected TBG ...C2z, C3z, and C2x, and the spinless antiunitary
time-reversal ..."
...
```

**Reasoning**

The Google Scholar results show relevant papers. In particular, result #1: "Twisted bilayer graphene. III. Interacting Hamiltonian and exact symmetries" by Bernevig, Song, Regnault, Lian (2021). This paper likely discusses the exact symmetries of the BM Hamiltonian, including $C_{2y}$ and $C_{3z}$. Let's open the first paper to see the symmetries.

**Action (Browser)**

```
url:https://journals.aps.org/prb/abstract/10.1103/PhysRevB.103.
205413
goal:Find the exact symmetries of the Bistritzer-MacDonald
Hamiltonian, particularly C2y and C3z, and whether they are
present when masses are different or tunneling has specific form.
```

**Observation (Browser)**

```
Evidence in page:
"We derive the explicit Hamiltonian of twisted bilayer graphene
(TBG) with Coulomb interaction projected into the flat bands and
study the symmetries of the Hamiltonian."
...
"We then prove that the interacting TBG Hamiltonian exhibits an
exact U(4) symmetry in the exactly flat band (nonchiral-flat)
limit."
```

**Reasoning**

The abstract didn't give the details. Let's try to access the paper directly or find a PDF. Maybe we can use the read-page tool on the PDF link or arXiv version. Let's search for the arXiv ID. The paper is Phys. Rev. B 103, 205413 (2021). It might be on arXiv.

**Action (Web Search)**

```
query:"Twisted bilayer graphene.  III. Interacting Hamiltonian and
exact symmetries" arXiv; "Twisted bilayer graphene.  III" "exact
symmetries"
```

**Observation (Web Search)**

```
Search results:
1.[2009.12376] Twisted bilayer graphene III. Interacting
Hamiltonian and exact symmetries.
url:https://arxiv.org/abs/2009.12376
...
4.arXiv:2009.12376v3 [cond-mat.str-el] 28 Apr 2022.
url:https://oar.princeton.edu/bitstream/88435/pr1x05xc8n/1/2009.
12376.pdf
...
```

**Reasoning**

Let's open the arXiv PDF to read the relevant sections about symmetries.

**Action (Browser)**

```
url:https://arxiv.org/pdf/2009.12376.pdf
goal:Find the discussion of exact symmetries of the
Bistritzer-MacDonald Hamiltonian, specifically time-reversal,
C2y, and C3z, and any conditions under which they are preserved
or broken.
```

**Observation (Browser)**

```
Evidence in page:
Appendix A: Review of the Single-particle Hamiltonian.
"Here we summarize the symmetries of TBG ...  Since graphene
has zero spin-orbit coupling (SOC), we can define a set of
spinless symmetries for TBG. In TBG, there are spinless unitary
discrete rotational symmetries C2z, C3z and C2x, and the spinless
anti-unitary time-reversal symmetry T, which satisfy [C3z, H0] =
[C2z, H0] = [C2x, H0] = [T, H0] = 0."
```

**Reasoning**

The retrieved passage gives the standard symmetry reference point: the continuum TBG Hamiltonian has spinless rotational symmetries including $C_{3z}$, and an anti-unitary time-reversal symmetry $T$. It does not by itself answer the present multiple-choice question, because this question adds layer-dependent masses and asks specifically about layer-inversion $C_{2y}$. Thus, the retrieved passage is useful for confirming the positive symmetries $\mathcal{T}$ and $C_{3z}$, but the absence of $C_{2y}$ is not taken from the source; it follows from the problem-specific unequal layer masses.
The Hamiltonian contains unequal layer masses $(m_{\mathfrak{b}}, m_{\mathfrak{t}}) = (0.65, 0.35)m_e$. Since layer-inversion $C_{2y}$ exchanges the two layers, it would require $m_{\mathfrak{b}} = m_{\mathfrak{t}}$; hence $C_{2y}$ is broken.
Time-reversal $\mathcal{T}$ maps $\tau \to -\tau$ and flips the spin, while sending $\mathbf{k} \to -\mathbf{k}$. The bottom kinetic term is unchanged. The top term $(\mathbf{k} - \tau\boldsymbol{\kappa})^2/(2m_{\mathfrak{t}})$ is also invariant because $(-\mathbf{k} + \tau\boldsymbol{\kappa})^2 = (\mathbf{k} - \tau\boldsymbol{\kappa})^2$.
...
Rotational $C_{3z}$ preserves each layer and cyclically permutes the three moire reciprocal vectors. Both the diagonal potential and the interlayer tunneling are constructed precisely from the three $C_{3z}$-related vectors, so the Hamiltonian is invariant under $C_{3z}$. The unequal masses do not affect $C_{3z}$ because it does not interchange the layers.
Hence the Hamiltonian manifests time-reversal and $C_{3z}$ symmetries, but not $C_{2y}$.
Final Answer: $\boxed{c}$

**Analysis**

This case illustrates an evidence-seeking CoI pattern. The model has an internal symmetry analysis gap, searches scholarly sources, follows a paper trail to the arXiv PDF, and then absorbs the retrieved symmetry statements into the final option-level decision.

**Case Study: Hybrid Closed-Loop Pattern in Chain-of-Interaction**

**Problem: (FrontierScience-Research #19, Score: 10.0/10.0)** How can a quantum shutter, pre- and post-selected in specific quantum states, simultaneously close an arbitrary number of slits to block a single photon in any state. How does a set of $K$ such shutters prevent the passage of $K$ photons?
You may use the following structure as a guide to solving the problem.
One-photon case:
1. Write down the initial (pre-selected) state of the shutter, in terms of the state of the shutter localized in slits $|i\rangle$. Write down the post-selected state of the shutter. 2. Write down the joint state of the shutter and photon at time $t$, the time of interaction between the shutter and the photon. 3. By post-selecting the joint state of the shutter and photon at time $t$, find the final state of the photon and show that it will only have reflected wave components, i.e. the screen operates as a perfect mirror.
Dual problem:
1. We have now $N$ shutters which close at least $N-1$ out of the $N$ slits. Nevertheless, we can pre- and post-select the state of these shutters in such a way that a single photon will see $N$ open slits. Again, derive the pre- and post-selected states of the shutter. Show that a single photon passes the slits without distortion, as if no shutters were present.

Generalization to $K$ photons:
1. In our method a single pre- and post-selected shutter closes $N$ slits for a single photon. What will happen if at time $t$ several photons are trying to pass through the slits? Start out by deriving the pre- and post-selection states for $K = 2$ and $K = 3$, then generalizing.

**Ground Truth:**

Relevant rubric items include:
Points: 0.25, Item: Writes the initial photon state $|\Psi_{\text{in}}\rangle_{\text{ph}} = \sum_i \alpha_i |i\rangle_{\text{ph}}$.
Points: 0.5, Item: Writes the pre-selected shutter state $|\Psi_1\rangle$.
Points: 0.5, Item: Writes the post-selected shutter state $|\Psi_2\rangle$.
Points: 1.5, Item: Writes the joint photon-shutter state after interaction.
Points: 0.25, Item: Explains that after post-selection the photon state has only reflected components $|\Psi_{\text{fin}}\rangle_{\text{ph}} = \sum_i \alpha_i |\tilde{i}\rangle_{\text{ph}}$, because transmitted shutter states are orthogonal to $|\Psi_2\rangle$.
Points: 1.0, Item: In the dual problem, writes the pre-selected shutter state $|\Phi_1\rangle$ and explains that $N$ shutters close at least $N-1$ slits while a single photon sees $N$ open slits.
Points: 1.0, Item: In the dual problem, writes the post-selected shutter state $|\Phi_2\rangle$ and shows that a single photon passes undistorted.
Points: 0.75, Item: For $K = 2$, writes the pre- and post-selected states of the first shutter.
Points: 0.75, Item: For $K = 2$, writes the pre- and post-selected states of the second shutter.
Points: 1.5, Item: For $K = 3$, writes the pre- and post-selected states of all three shutters.
Points: 2.0, Item: For general $K$, writes the general pre- and post-selected shutter states $|\Psi_1^K\rangle$ and $|\Psi_2^K\rangle$.

**Reasoning**

We need to answer a physics problem about a quantum shutter, pre- and post-selected, that can close all $N$ slits simultaneously for a photon, and how a set of $K$ shutters prevents passage of $K$ photons. Also a dual problem where $N$ shutters close at least $N-1$ slits but a single photon sees $N$ open slits. We need to derive pre- and post-selected states, joint states, show reflection, etc. This is a known problem in quantum foundations. Let's search.

**Action (Web Search)**

```
query:quantum shutter pre- and post-selected close all N slits
simultaneously photon; Aharonov Vaidman quantum shutter N slits
preselected postselected; K shutters prevent passage of K photons
quantum shutter problem
```

**Observation (Web Search)**

```
Search Results:
1.[quant-ph/0206074] How One Shutter Can Close N Slits - arXiv
url:https://arxiv.org/abs/quant-ph/0206074
snippet:It is shown that a quantum shutter, pre- and post-selected
in particular quantum states, can close simultaneously arbitrary
number of slits...
...
4.How one shutter can close N slits | Phys. Rev. A - APS
Journals
url:https://link.aps.org/doi/10.1103/PhysRevA.67.042107
snippet:A set of K preselected and postselected shutters can close
the slits preventing the passage of K or less photons.
```

**Reasoning**

We have found the relevant paper: "How One Shutter Can Close N Slits" by Aharonov and Vaidman. Let's open the PDF or arXiv to get the exact derivations. We need to answer the problem in detail: one-photon case, dual problem, generalization to $K$ photons.

**Action (Browser)**

```
url:https://arxiv.org/pdf/quant-ph/0206074.pdf
goal:Extract the full derivations, pre- and post-selected states,
joint states, and proofs for the one-photon case, the dual
problem, and generalization to K photons. Include all equations
and reasoning.
```

**Observation (Browser)**

Evidence in page:
"Our task is to close all $N$ slits at time $t$ with this single shutter."
...
"To achieve this task we prepare (pre-select) the shutter at $t_1$ in the state"

$$|\Psi_1\rangle = \frac{1}{\sqrt{2N-1}}\left(\sum_{i=1}^{N}|i\rangle + \sqrt{N-1}|N+1\rangle\right).$$

...
"We post-select the shutter at $t_2$ in the state"

$$|\Psi_2\rangle = \frac{1}{\sqrt{2N-1}}\left(\sum_{i=1}^{N}|i\rangle - \sqrt{N-1}|N+1\rangle\right).$$

**Reasoning**

The read-page output gave us the key equations and derivations from the paper. We should also verify some details, especially the regrouped joint state, the normalization, and the orthogonality condition that removes the transmitted photon components after post-selection.

**Action (Browser)**

```
url:https://arxiv.org/pdf/quant-ph/0206074.pdf
goal:Extract equation (4) exactly as printed in the PDF, including
the coefficient of the |N+1> term.  Also check the normalization
of the pre- and post-selected states and the orthogonality
condition.
```

**Observation (Browser)**

Evidence in page:

$$|\Psi\rangle_{s,ph} = \frac{1}{\sqrt{2N-1}} \sum_{i=1}^{N} \alpha_i |i\rangle|\tilde{i}\rangle_{\text{ph}} + \frac{1}{\sqrt{2N-1}} \sum_{j=1}^{N} \alpha_j \left( \sum_{i \neq j} |i\rangle + \sqrt{N-1}|N+1\rangle \right) |j\rangle_{\text{ph}}.$$

...

"We can see that all states of the shutter appearing in the second term in the last expression, i.e. all states correlated with a photon which passed through the screen, are orthogonal to the post-selected state $|\Psi_2\rangle$. Therefore, after the post-selection, the photon state will have only reflected wave components."

**Reasoning**

The page clarifies the exact form of the joint state. The transmitted branch associated with photon path $j$ contains the shutter state

$$\sum_{i \neq j} |i\rangle + \sqrt{N-1}|N+1\rangle.$$

Its overlap with $|\Psi_2\rangle$ should vanish. For the $K$-photon generalization, I want to verify the more general overlap formula symbolically: for a shutter indexed by $m$ and $D$ occupied distinct slits, the transmitted shutter state should be orthogonal exactly when $m = D$.

**Action (Code Executor)**

```
import sympy as sp

N, m, D = sp.symbols('N m D', positive=True)
# Post-selected shutter m has denominator sqrt(2N-m)
# Transmission state for D distinct occupied slits has N-D
unoccupied slit terms
# Inner product = (N-D)/(2N-m) - (N-m)/(2N-m)
inner = (m - D) / (2*N - m)

# K=2 front shutter:  m=2, two occupied slits D=2
sp.simplify(inner.subs({m:  2, D: 2}))
```

**Observation (Code Executor)**

```
0
```

**Reasoning**

Good. The symbolic check confirms that the front shutter in the $K = 2$ construction removes the branch in which two photons pass through two distinct slits. Let's also keep the general expression, because it explains the induction step: a shutter with index $m$ blocks exactly the branch with $D = m$ occupied slits.

**Action (Code Executor)**

```
import sympy as sp
N, m, D = sp.symbols('N m D', positive=True)
inner = (m - D) / (2*N - m)
sp.simplify(inner)
```

**Observation (Code Executor)**

```
(-D + m)/(2*N - m)
```

## Reasoning

The verification gives

$$\langle \Psi_2^{(m)} | T_D \rangle = \frac{m-D}{2N-m}.$$

Thus the overlap vanishes when $m = D$. This is the closed-loop check needed for the $K$-photon construction: the first shutter, indexed by $K$, eliminates branches with $K$ distinct occupied slits; after at least one photon is reflected, the surviving photons occupy at most $K-1$ distinct slits, and the next shutter handles $m = K-1$, continuing by induction.
Final Answer:
The incoming photon is $|\Psi_{\text{in}}\rangle_{\text{ph}} = \sum_{i=1}^{N} \alpha_i |i\rangle_{\text{ph}}$. Choose

$$|\Psi_1\rangle = \frac{\sum_i |i\rangle + \sqrt{N-1}|N+1\rangle}{\sqrt{2N-1}}, \qquad |\Psi_2\rangle = \frac{\sum_i |i\rangle - \sqrt{N-1}|N+1\rangle}{\sqrt{2N-1}}.$$

After interaction,

$$|\Psi\rangle_{s,\text{ph}} = \frac{1}{\sqrt{2N-1}} \sum_i \alpha_i |i\rangle |\tilde{i}\rangle_{\text{ph}} + \frac{1}{\sqrt{2N-1}} \sum_j \alpha_j \left( \sum_{i \neq j} |i\rangle + \sqrt{N-1}|N+1\rangle \right) |j\rangle_{\text{ph}}.$$

The shutter state correlated with each transmitted component is orthogonal to $|\Psi_2\rangle$, so post-selection removes all transmitted components and gives $|\Psi_{\text{fin}}\rangle_{\text{ph}} = \sum_i \alpha_i |\tilde{i}\rangle_{\text{ph}}$.
For the dual problem, let $|C_i\rangle = |\text{op}\rangle_i \prod_{j \neq i} |\text{cl}\rangle_j$ and $|C_{\text{all}}\rangle = \prod_j |\text{cl}\rangle_j$. Take

$$|\Phi_1\rangle = \frac{\sum_i |C_i\rangle + \sqrt{N-1}|C_{\text{all}}\rangle}{\sqrt{2N-1}}, \qquad |\Phi_2\rangle = \frac{\sum_i |C_i\rangle - \sqrt{N-1}|C_{\text{all}}\rangle}{\sqrt{2N-1}}.$$

Although each branch closes at least $N-1$ slits, post-selection cancels the reflected components, so the photon passes undistorted: $|\Psi_{\text{fin}}\rangle_{\text{ph}} \propto \sum_i \alpha_i |i\rangle_{\text{ph}}$.
For $K$ photons, use $K$ shutters in series. The $m$-th shutter is

$$|\Psi_{1,2}^{(m)}\rangle = \frac{\sum_i |i\rangle \pm \sqrt{N-m}|N+1\rangle}{\sqrt{2N-m}}, \qquad m = K, K-1, \ldots, 1,$$

where $+$ gives the pre-selected state and $-$ gives the post-selected state. Thus $K = 2$ uses $m = 2, 1$, $K = 3$ uses $m = 3, 2, 1$, and the same construction gives the general $K$-photon result. Hence $K$ pre- and post-selected shutters close $N$ slits for any state of $K$ or fewer photons.

## Analysis

This case illustrates a hybrid closed-loop pattern from a full-credit FrontierScience-Research rollout. The model retrieves the original quantum-shutter paper, grounds the pre-/post-selected state construction in the paper's equations, then uses symbolic execution to verify the orthogonality condition that makes the shutter block transmitted branches. The final answer absorbs both the external derivation and the symbolic check to give the one-photon, dual, and general $K$-photon constructions.